Leakage, Shift and the Operating Point: Evaluating Blood-Smear Leukemia Classifiers Across Time and Laboratories

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***Abstract*—Blood-smear leukemia classifiers often report 97–100% accuracy, yet few are used in practice. We separated three questions—discrimination, fixed-threshold performance and review workload—in 106 patients with acute myeloid leukemia (34 with acute promyelocytic leukemia, APL) split by diagnosis date, C-NMC 2019 (73 subjects; lymphoblastic leukemia versus normal B cells), and a prespecified evaluation on an independent cohort from another laboratory (24 APL, 105 other AML). Image-level cross-validation (CV) yielded up to 95% accuracy; new patients, 69–90%, depending on task and model. For the task in which each cell inherited its patient's diagnosis, image-level CV inflated the area under the ROC curve (AUROC) by 0.04–0.11 relative to patient-grouped CV, and the external generalization gap to the later APL cohort was a further 0.17–0.24; an acquisition-order proxy was strongly associated with APL in the development cohort, suggesting a possible scan-time confounder. A prespecified pipeline trained on 82 patients ranked APL patients in the independent laboratory with an AUROC of 0.86 (95% CI 0.77–0.94); a hematology foundation model plus engineered features reached 0.88. At the development threshold, all 24 APL patients were detected at a specificity of 0.00. Referring the top-ranked 20% of patients captured 15–17 of 24 APL patients; capturing 95% required reviewing 60–75%. In an exploratory reverse-transfer analysis, pooled-feature models reached AUROCs of 0.75–0.78, attention-based multiple instance learning did not transfer, and thresholds again failed. Patient-grouped and external validation are necessary but not sufficient; these models are candidates for prioritization research, not validated screening tools.**

***Index Terms*—Acute promyelocytic leukemia, acute lymphoblastic leukemia, blood smear, data leakage, dataset shift, deep learning, external validation, multiple instance learning, sensitivity.**

I. Introduction

ACUTE leukemia is usually first suspected on a peripheral blood smear [1]. Acute promyelocytic leukemia (APL) shows how much rides on that first look: patients can bleed to death within days unless all-trans retinoic acid is started as soon as APL is suspected, often before genetic confirmation arrives. Reading smears takes experience that is not available everywhere, which is why automated smear analysis has attracted so much work.

Many deep-learning models report 97–100% accuracy for leukemia detection from single-cell images [2], [3], and weakly supervised models reach AUCs above 0.95 for APL [4], [5]. Yet few of these systems are used in practice. Part of the gap may lie in how they are evaluated. When cells from the same patient appear in both training and test sets, a model can succeed by recognizing the patient—the stain, the smear, the day it was scanned—rather than the disease; when the label is the patient's diagnosis copied onto every cell, this shortcut is always available [6], [7]. And accuracy alone hides the error that matters most clinically: a sick patient called healthy. Similar problems have been documented across machine-learning-based science [8] and medical imaging [9]–[11].

We measure these effects on a public cohort of 106 patients with acute myeloid leukemia (AML), 34 with APL, repeat the analysis on the C-NMC 2019 dataset of acute lymphoblastic leukemia (ALL), and evaluate the locked pipelines, under a plan committed before scoring, on an independent cohort from another laboratory. Our contributions are:

- C1 *Leakage separated from shift.* With identical data and models, we compare image-level cross-validation (CV), patient-grouped CV and an external set, and split the gap into a leakage part and an external generalization gap, for labels that belong to the patient and labels that belong to the cell; we also identify a scan-time confounder.
- C2 *Three kinds of external test.* A temporal split of the APL-AML cohort, a held-out subject fold of C-NMC, and a prespecified evaluation on 129 patients from an independent laboratory, complemented by pooled CV over all 106 APL-AML patients.
- C3 *Sensitivity-first reporting.* Thresholds are fixed on development data for a recall of at least 0.95; we report discrimination, fixed-threshold performance (missed patients, specificity, F1 score, negative predictive value (NPV)) and review workload separately, together with calibration, net benefit, and what local cases can and cannot achieve when a new laboratory adapts the threshold. We distinguish probability calibration (do predicted risks match observed ones?), threshold adaptation (moving the decision threshold to meet an operating target) and local validation (testing the resulting rule on patients not used to set it).
- C4 *Design decisions driven by the data.* An exploratory analysis and a label-free shift test decide the preprocessing; engineered, ImageNet, fine-tuned and hematology-foundation-model features, principal component analysis

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(PCA) and early stopping are each reported before and after.

C5 *Pre-specified pipelines with their cost.* A primary and a low-complexity secondary pipeline, both trainable on a CPU, are reported with storage, model size and time per patient.

II. Related Work

Table I summarizes representative studies and the open question each leaves, linked to the contribution of this paper that addresses it.

Cell-level classification. The C-NMC 2019 challenge [12] held out whole subjects for testing; ResNeXt models reached a weighted F1 score of 0.889 [13] and a five-CNN ensemble 0.886 [14]. Transformer and hybrid designs report 0.97–1.00 accuracy on C-NMC and ALL-IDB [2], [3], [15], and CNN features with classical classifiers reach 0.88 on three classes [16]. Because the split unit is often not reported, and some studies combine sources in which class and source coincide, it remains open how much of this carries over to new patients. A subject-disjoint benchmark on C-NMC found an AUROC of 0.913 [17], for a single cell-level label.

Patient-level diagnosis. Expert-level single-cell classification [18], [19] enabled multiple instance learning (MIL) for genetic AML subtypes, including APL, in a cohort from the Munich Leukemia Laboratory [20] that we use here as an independent test set and mutation status [21], [22]. Sidhom *et al.* [23] released the APL-AML cohort used here, validated their model on the later patients and compared it with ten clinicians; Manescu *et al.* [4] reported an AUC of 0.96 on the same patients under random three-fold CV, and Eckardt *et al.* detected APL in bone-marrow smears [5]. Engineered morphometry has held up in an external cohort for myelodysplasia [24]. The number of missed patients at a pre-set operating point, and calibration, are rarely reported.

Foundation models and shift. Hematology foundation models such as DinoBloom [25], built on DINOv2 [26], and their successors [27], [28] aim at robustness to scanners and stains; LeukemiaAttri images the same films with different microscopes [29], and a DinoBloom-based transformer developed on data collected from 6,610 individuals of one diagnostic laboratory (a smaller subset was used for training after exclusions) was validated externally, including on our cohort [30]. In histopathology, stain normalization and augmentation trade off differently across shifts [31].

III. Materials and Methods

A. Datasets

APL-AML. The public cohort of Sidhom *et al.* [23] contains single-cell images from Wright-stained peripheral blood smears of 106 adults with AML (34 APL, 72 other subtypes) treated at Johns Hopkins Hospital between 2010 and 2020 and captured with a CellaVision DM100 analyzer. The authors split it by date: 82 patients diagnosed before January 2019 form the Discovery cohort and 24 diagnosed later the Validation cohort. Of 25,724 images (about 360×363 pixels), 15,632 carry an operator-confirmed analyzer cell type (Table II; examples in Fig. 1; composition in Fig. S1). Ten files (0.04%) are truncated and unreadable; they are kept as missing rows and imputed inside each training fold, and 54 are full microscope fields rather than single-cell crops. APL makes up 27% of Discovery and 50% of Validation patients, and the number of cells per patient is confounded with the diagnosis in opposite directions in the two cohorts. The original authors note that all Discovery smears scanned before 2018 were APL [23]; we examine this in Section IV-C.

C-NMC 2019. To test whether the findings hold in another laboratory, we used C-NMC 2019 [12], released by The Cancer Imaging Archive [32] and obtained from a Kaggle mirror whose file and subject counts match the official training set. It contains 10,661 segmented cells (450×450 pixels, black background) from 73 subjects: 47 with B-lineage ALL and 26 healthy individuals whose normal B cells serve as controls (Table S3; Fig. 1). Before any analysis we designated official folds 0–1 as development set (42 subjects, 7,108 cells) and fold 2 as external set (31 subjects, 17 with ALL, 3,553 cells). Every cell carries its subject's diagnosis; unlike APL-AML, the ALL cells were selected as malignant lymphoblasts, so the label is also partly a property of the cell.

Independent APL cohort (MLL). To test the final pipelines on patients from another institution, we used the AML-Cytomorphology_MLL_Helmholtz collection of Hehr *et al.* [20] (The Cancer Imaging Archive, CC BY 4.0; obtained from a Kaggle mirror with one folder per patient). It contains single-cell images (144×144 pixels, Metafer scanner, $40\times$) from peripheral blood smears of 189 patients of the Munich Leukemia Laboratory diagnosed 2009–2020. We used the 24 patients with APL (PML::RARA) as positives and the 105 patients with AML with NPM1 mutation, CBFB::MYH11 or RUNX1::RUNX1T1 as negatives, with up to 200 randomly drawn cells each (25,800 cells); the 60 healthy donors were excluded. The cohort was never used for training, selection, thresholds or calibration, and its analysis plan was committed to the code repository before any model was scored on it (commit `acf173b`).

B. Learning Tasks

Five tasks are defined (Table III). In T1 every APL-AML cell inherits its patient's diagnosis, a common way of building single-cell datasets; this is a weak label, not an expert judgment of the cell's own morphology, and T1 serves to expose leakage rather than as a cell-level disease classifier. In T2 the label is the cell's own type (blast versus other labeled white cells). T3 asks whether a patient has APL. T4 and T5 are the C-NMC counterparts of T1 and T3.

C. Preprocessing, Segmentation and Stain Normalization

Fig. 2 outlines the pipeline. A manifest records every image with its patient, cohort, diagnosis and cell label; no image is removed because of its label. For the engineered features, each image is resized to 192×192 pixels and separated into hematoxylin, eosin and residual channels by color deconvolution [33]. The nucleus is segmented by Otsu thresholding [34] of

TABLE I
REPRESENTATIVE PREVIOUS STUDIES (RESULTS AS REPORTED; NOT COMPARABLE ACROSS DATASETS OR TARGETS) AND THE OPEN QUESTION EACH LEAVES, WITH THE CONTRIBUTION OF THIS PAPER (C1–C5) THAT ADDRESSES IT.

Study	Data	Evaluation	Reported	Open question (C)
Prellberg & Kramer 2019 [13]	C-NMC cells	Unseen subjects	F1 0.889	Missed cases, calibration (C3)
Mondal *et al.* 2021 [14]	C-NMC cells	Unseen subjects	AUC 0.941	Operating point, cost (C3, C5)
Kasim *et al.* 2025 [16]	390 cells, 2 sources	Image-level split	Acc. 0.88	Class = source (C1, C2)
Jammal *et al.* 2025 [2], [3]	C-NMC, ALL-IDB	Split unit unclear	Acc. 0.97–1.00	New patients? (C1, C2)
Albzour 2026 [17]	C-NMC cells	Subject-disjoint	AUROC 0.913	Leakage vs. gap (C1)
Hehr *et al.* 2023 [20]	189 patients, AML subtypes	Patient CV	Subtyping	External site, threshold (C2, C3)
Sidhom *et al.* 2021 [23]	APL-AML (*this cohort*)	Temporal; reader study	AUC 0.743	Missed patients (C3)
Manescu *et al.* 2023 [4]	APL-AML and others	Random 3-fold CV	AUC 0.96	Temporal split (C1, C2)
de Almeida *et al.* 2023 [24]	352 + 63 slides	External cohort	AUC $\leq$0.90	Acute leukemia (C4)
Koch *et al.* 2024 [25]	380k cells, 13 sets	Probing, MIL	Beats generic	Patient-level external (C2)
Dasdelen *et al.* 2026 [30]	One laboratory, 6,610 individuals	Two external sets	Malignancy AUC 0.97	APL vs. AML, threshold (C3, C5)

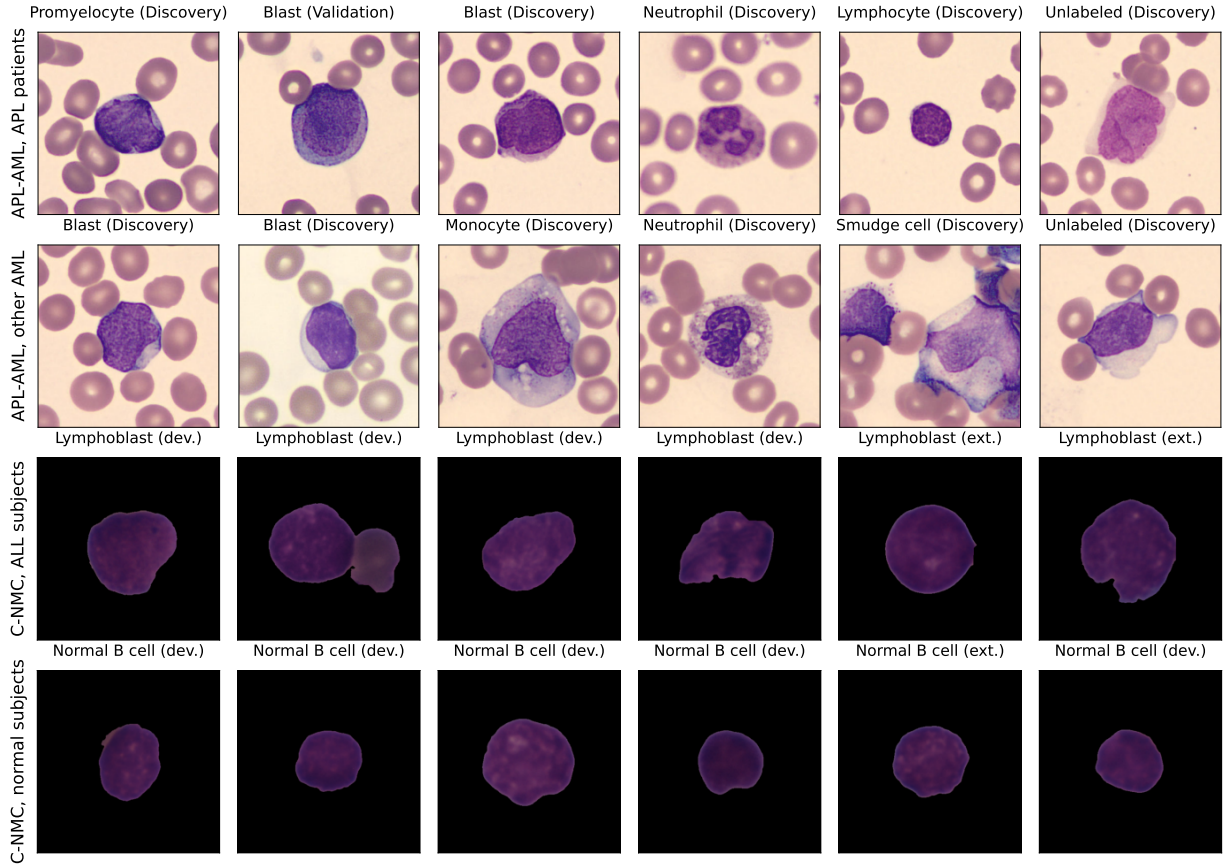


Fig. 1. Sample images from both datasets (randomly drawn, with cell type and cohort). Rows 1–2: APL-AML single-cell crops from Wright-stained peripheral blood smears (CellaVision DM100), with neighboring red cells in the field; cell types are the operator-confirmed analyzer labels, and about 39% of cells are unlabeled. Rows 3–4: C-NMC 2019 segmented B-lineage cells on a black background, labeled only by the subject's diagnosis (dev.: folds 0–1; ext.: fold 2). The datasets differ in stain, magnification, background and cell selection.

the smoothed hematoxylin channel, followed by morphological opening and selection of the component nearest the image center. Pale cytoplasm could not be separated reliably from neighboring red cells, so we describe it with two proxies: a basophilic region of blue-dominant pixels within one nuclear radius and a perinuclear ring 2–9 pixels wide (Fig. S5). For C-NMC, whose black background would be read as dense chromatin, pixels whose brightest channel is below 15 were

TABLE II
COMPOSITION OF THE APL-AML COHORT.

Cohort	Diagnosis	Patients	Cells	Labeled cells	Cells per patient median (range)
Discovery	AML	60	11,523	8,077	134 (98–984)
	APL	22	8,764	4,082	290 (112–1,066)
Validation	AML	12	3,589	1,893	252 (116–554)
	APL	12	1,848	1,580	140 (83–268)
Total		106	25,724	15,632	152 (83–1,066)

TABLE III
LEARNING TASKS (DEVELOPMENT / EXTERNAL SET). T1–T3: APL-AML (DISCOVERY / VALIDATION); T4, T5: C-NMC 2019 (FOLDS 0–1 / FOLD 2).

Task	Unit	Positive class	Size
T1 Diagnosis on cells	Cell	Cell from an APL patient (43.2%)	20,287 / 5,437 cells
T2 Blast detection	Cell	Labeled blast (28.0%)	9,694 / 2,863 cells
T3 Patient diagnosis	Patient	APL (22 of 82)	82 / 24 patients
T4 C-NMC cells	Cell	Cell from an ALL subject (67.7%)	7,108 / 3,553 cells
T5 C-NMC subjects	Subject	ALL (30 of 42)	42 / 31 subjects

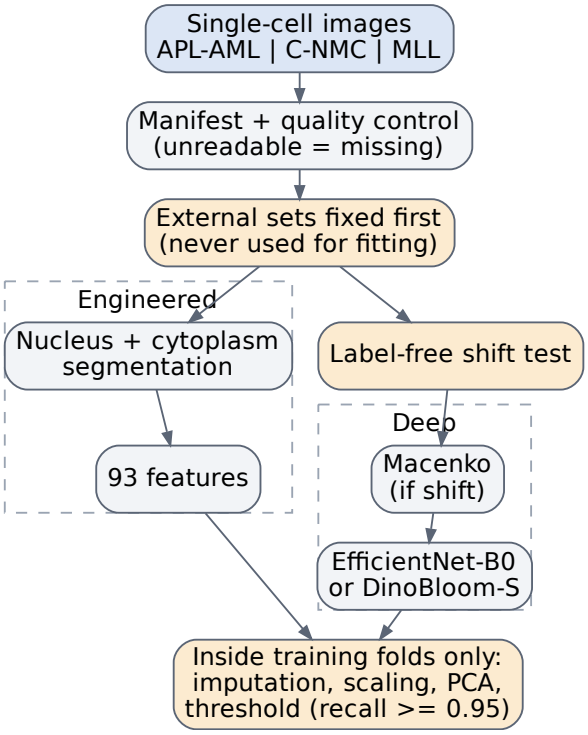


Fig. 2. Preprocessing for both datasets. Orange steps prevent leakage: the external sets are fixed before any analysis, and every statistic—imputation, scaling, PCA, class weights and the decision threshold—is fitted inside the training fold.

set to white before engineered features were extracted; the CNN received the original images.

For the CNN we compared no preprocessing, edge sharpening (unsharp mask), contrast-limited adaptive histogram equalization (CLAHE), Reinhard color normalization [35] and Macenko stain normalization [36], with a rule fixed in advance: the method had to reduce the measurable cohort shift without lowering external performance. Macenko normalization estimates the two stain directions of each image in optical-density space and maps them, and their concentrations, onto reference stains estimated from 300 Discovery images. Because smear backgrounds are beige rather than white, the illumination of each image is taken from the 99th percentile of each color channel. Normalization is applied only when a label-free shift test is positive: a classifier assigns control-class cells (AML cells, normal B cells) to the development or external set with patient-grouped 5-fold CV, and a permutation test that shuffles the set labels between patients gives the p-value. The test reads the unlabeled external images but none of their diagnoses; it is transductive in the sense that a new site would run it on its own unlabeled images before deployment. Engineered features were always computed on the original images, because their color features are diagnostic in APL.

D. Feature Representations

The 93 engineered features describe nuclear shape (9), the basophilic region and nucleus-to-region ratio (10), stain and color in the nucleus and the ring (26), chromatin texture from gray-level co-occurrence matrices [37] and nuclear entropy (16), uniform local binary patterns [38] (28), and pale nuclear spots and dark granules (4). As a deep representation we used a frozen EfficientNet-B0 [39] pretrained on ImageNet (1,280 dimensions per cell); the fusion representation concatenates both after column-wise standardization. Because a frozen generic encoder may understate deep learning, we also fine-tuned EfficientNet-B0. Full fine-tuning was not feasible on our CPU-only workstation, so we froze the stem and first five stages, cached their output, and trained the last two stages, the head and a new classifier with AdamW [40] (learning rate 3×10^{-4}, cosine schedule, 6 epochs, best epoch on 15% of the training patients, up to 5,000 cells per fold, one seed; Table S1). As a hematology foundation model we used DinoBloom-S [25], a ViT-S/14 trained with DINOv2 [26] on more than 380,000 blood and bone-marrow cells; its weights were taken unchanged from the authors' release, and the 384-dimensional class token of each 224×224 cell image (original colors) was used as a frozen representation, alone or concatenated with the engineered features.

On the frozen embeddings we tested PCA, fitted on the training cells of each fold. The embeddings are standardized and projected onto the leading k eigenvectors $\mathbf{W}_k$ of their covariance [41], $\mathbf{z} = \mathbf{W}_k^\top \tilde{\mathbf{x}}$. We pre-specified $k \in \{16, 32, 64\}$ and added $k \in \{4, 8, 128, 256\}$ as a sensitivity analysis. A patient vector holds five statistics per dimension, so PCA reduces it from 6,400 to $5k$ values. The pooled component scores are centered and divided by one common scale ("natural scale"), so that the components keep their relative variances; standardizing each component separately would whiten them and give noise directions the same weight as the leading ones. Our first analysis used whitening; every PCA result in this paper uses the natural scale unless labeled otherwise. Further feature-engineering steps (correlation filter, per-slide stain normalization of features, L1 selection, a predicted differential count) are described in Table S1 and reported in Table S5.

E. Classification Models

Fig. S3 shows how the components fit together. At cell level we used L2-regularized logistic regression (LR) and LightGBM [42] with class-balanced weights, each as a single pipeline that imputes and scales with training data only. For patients and subjects (T3, T5), up to 500 randomly sampled cells represent each patient. The pooled model summarizes every feature by its mean, standard deviation and 10th, 50th and 90th percentiles and applies LR. Gated attention MIL [43] maps each cell to $\mathbf{h}_k = \mathrm{ReLU}(\mathbf{W}\mathbf{x}_k)$ (128 units), weights the cells by

$$a_k = \frac{\exp\{\mathbf{w}^\top(\tanh(\mathbf{V}\mathbf{h}_k) \odot \sigma(\mathbf{U}\mathbf{h}_k))\}}{\sum_j \exp\{\mathbf{w}^\top(\tanh(\mathbf{V}\mathbf{h}_j) \odot \sigma(\mathbf{U}\mathbf{h}_j))\}}, \qquad (1)$$

and classifies $\mathbf{z} = \sum_k a_k \mathbf{h}_k$; because the weights sum to one, the output does not depend on the number of cells. It is trained for 40 epochs with AdamW (learning rate 2×10^{-4}, weight decay 10^{-3}), class-weighted cross-entropy and random removal of 20% of cells. The fine-tuned CNN gives a patient score as the mean of its cell probabilities.

F. Final Pipelines and Pre-Specification

Before the final analysis we fixed one primary and one secondary pipeline (Fig. 3) and committed the specification to the code repository (commit `f47f3ec`, 21 October 2026, 07:19 UTC+3). The primary pipeline normalizes stains when the shift test is positive, scores every cell with the fine-tuned CNN, and averages the percentile scores of three patient-level members—the mean fine-tuned cell probability, engineered features with pooled LR, and engineered features with attention MIL—with no weights fitted. The secondary pipeline was declared as a low-complexity option: stain normalization, the frozen CNN, PCA with 64 components, pooled statistics and LR. Both use a threshold fixed on development out-of-fold scores for a recall of at least 0.95. Their intended use is to prioritize smears for urgent review and confirmatory testing, not to rule APL out.

What the pre-specification does and does not protect. The APL-AML Validation cohort was never used to fit a model, threshold or calibration. It was, however, not a single-use test set. In our first analysis (20 October 2026) and during this revision we evaluated many configurations on it—feature-engineering steps, PCA sizes, preprocessing variants on a subset—and those numbers were known when later choices were made, including the pre-specification. The primary pipeline was assembled from the components that had been most consistent across both datasets, but we cannot exclude that knowledge of Validation results shaped it. The APL-AML Validation numbers should therefore be read as those of a reused test set; the C-NMC external fold (also evaluated for several configurations, but never used for selection), the pooled analysis and the decomposition into leakage and external generalization gap provide the more protected evidence, and the independent MLL cohort, analyzed once under a plan committed beforehand, provides an untouched test. Table S14 lists, for every analysis, whether it is confirmatory or exploratory and what its test set had been exposed to. For that test we also pre-declared one model added in this revision, DinoBloom-S combined with engineered features and pooled LR, chosen because it had the highest development CV on C-NMC before the independent cohort was scored.

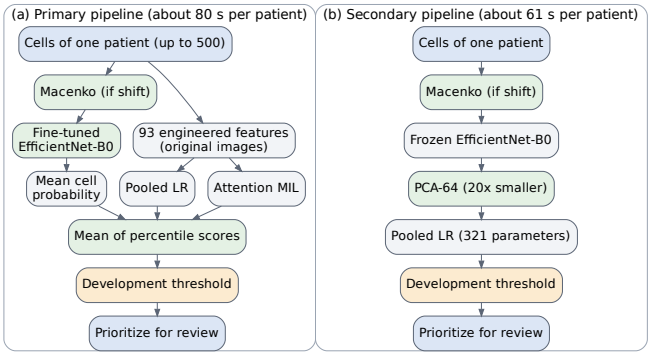


Fig. 3. End-to-end pipelines from the single-cell images of a patient to the decision. Orange: fixed on development data; green: components retained because they improved performance (primary) or reduced complexity (secondary). Measured cost per cell on 2 vCPUs: Macenko normalization 0.071 s, CNN 0.051 s, engineered features 0.038 s (500 cells per patient); settings in Table S1.

G. Evaluation Protocol and Metrics

Every cell-level model was evaluated in three ways (Fig. S4): 5-fold CV on the development set with image-level folds, as is common in the literature; 5-fold CV with patient-grouped folds; and the external set. We define *leakage* as image-level minus patient-grouped CV AUROC (same data, only the split differs) and the *external generalization gap* as patient-grouped CV minus external AUROC (new patients). The second quantity is descriptive: a later or independent cohort changes stain, scanner, patient mix, selection and label distribution at once, and the gap does not isolate any one of them. For brevity we call it the gap. We also scored the external set with the models trained under image-level folds, to see whether leaky training harms the model or only the estimate. Within each training fold, an inner 80/20 split of the same kind fitted a calibration temperature [44]. External predictions of patient-level models average all fold models over three seeds.

Since missing an APL patient can delay life-saving treatment, whereas a false alarm leads to a confirmatory genetic test, every decision threshold was fixed in advance on development out-of-fold scores to reach a recall of at least 0.95. At that operating point we report recall with the number of missed positives, specificity, F1 score, NPV and accuracy; AUROC as a threshold-free summary; and calibration as the Brier score, calibration intercept and slope [45] and expected calibration error (ECE). Confidence intervals (95%) come from bootstrap resampling of patients [46]; differences between models use a paired patient-level bootstrap, with Holm correction [47] when several comparisons share a reference. C-NMC was analyzed with the same protocol.

Three further analyses address what the external tests alone cannot. First, because 24 external patients give wide intervals [48], [49], we repeated the main patient-level comparisons

with three repeats of 5-fold CV over all 106 APL-AML patients, stratified by diagnosis and cohort, with thresholds chosen by inner CV. Second, we simulated local threshold adaptation: k confirmed local patients ($k/2$ APL, $k = 2, \ldots, 8$) are drawn from the Validation cohort, the threshold is lowered if needed to just below the lowest local APL score, and the remaining $24 - k$ patients are scored, over 2,000 draws. Third, decision-curve analysis [50] estimated the net benefit at an APL prevalence of 10%, after logistic recalibration on Discovery and a shift to the target prevalence.

H. Implementation

All experiments ran on a CPU-only workstation (2 vCPUs, 7.8 GB RAM; Table S2). Table S1 lists every setting; settings were fixed in advance or taken from library defaults, and none was tuned on an external set. Reporting follows TRIPOD+AI [51] (checklist in Table S16).

IV. Results

A. Exploratory Analysis and the Shift Test

Both datasets are imbalanced in different ways: APL makes up 27% of Discovery and 50% of Validation patients; ALL makes up 71% of C-NMC development and 55% of external subjects. We therefore used class-balanced weights, did not resample cells, set thresholds for the positive class, and report recall, specificity, missed cases, F1 score and AUROC rather than accuracy alone.

Fig. 4 summarizes the APL-AML exploratory analysis. Every feature group shifted between cohorts, stain and color most (median standardized mean difference 0.34). Among AML patients, a classifier separated Validation from Discovery cells with an AUROC of 0.753 on stain and color features (permutation $p = 0.005$), 0.704 on all engineered features ($p = 0.010$) and 0.583 on frozen CNN embeddings ($p = 0.039$; Table S4). Thirty-seven feature pairs were nearly redundant ($|r| > 0.9$), and single-feature patient-level AUROCs in the two cohorts correlated only moderately ($r = 0.38$). C-NMC looked different (Fig. S2): feature shifts were small (median 0.04–0.16 per group) and normal B cells could not be assigned to their set (AUROC 0.44–0.47 on the full data, $p \geq 0.45$ for every representation; 0.52–0.56 on the preprocessing subset, Table VII). By the pre-stated rule, stain normalization was therefore applied to APL-AML only. On the full data, however, CNN embeddings of normalized images were slightly *more* identifiable than before (0.603 vs. 0.583, $p = 0.06$), whereas on the subset used for the selection normalization had lowered identifiability from 0.575 to 0.543. Macenko normalization reduces color differences but does not remove the acquisition signatures the CNN can see, and its selection rested on a subset that included Validation patients; Section IV-H therefore reports the primary pipeline with and without it.

B. Leakage Versus Shift

Table IV and Fig. S6 separate the two sources of optimism. When each cell carried its patient's diagnosis (T1), image-level CV reported AUROCs of 0.938–0.987, patient-grouped CV 0.867–0.901 and the external cohort 0.658–0.734. Leakage accounted for 0.04–0.11 and the external generalization gap for 0.17–0.24; both components were positive in every patient-level bootstrap interval (leakage lower bounds 0.02–0.07, gap lower bounds 0.05–0.07), although the gap intervals are wide (upper bounds 0.31–0.43) because only 24 patients form the external cohort. When the label described the cell itself (T2), leakage was 0.01–0.03 (intervals excluding zero) and the gap indistinguishable from zero. On C-NMC cells (T4), leakage (0.07–0.08, intervals excluding zero) was larger than the gap (0.02–0.06, intervals including zero), consistent with a random subject-disjoint fold of one collection. Leakage is thus largest when the label belongs to the patient, as expected, but on the temporally split APL-AML cohort the gap between periods was the larger part of the overstatement.

Leaky training did not produce worse models. Classifiers trained under image-level folds ranked external cells as well as or slightly better than those trained under grouped folds (T1: 0.677–0.760 vs. 0.658–0.734), presumably because each fold model saw cells of every patient. What leakage corrupted was the operating point: a threshold tuned for a recall of 0.95 on leaky folds recovered only 32–66% of the cells of new APL patients, against 69–81% for thresholds tuned on grouped folds. Table V shows the result in the terms most studies report: the same models reached 0.88–0.95 accuracy under image-level CV at a 0.5 threshold, but 0.69–0.73 on new APL-AML patients, 0.87–0.90 for blasts and 0.82–0.83 on C-NMC.

C. Scan Time as a Confounder

The images carry no scan dates, but the analyzer numbers the cells it signs sequentially, so the median cell number of a patient is a time proxy. It separated Validation from Discovery patients almost perfectly (AUROC 0.992), which supports its validity. Within Discovery, it predicted APL with an AUROC of 0.775 (Table VI). In the Validation cohort it carried no diagnostic information (0.479), so the confounding is confined to the development data. The images themselves record scan time: among AML patients only, pooled features separated early from late scans with AUROCs of 0.74–0.79 for every representation, highest for all engineered features (0.794) and including shape and texture (0.767); Macenko normalization lowered this for color features from 0.760 to 0.658 but did not remove it. To see whether the diagnostic models depend on it, we repeated the Discovery CV in the 31 patients (18 APL) scanned while both diagnoses were being scanned, where scan time was no longer positively associated with APL (AUROC 0.325, i.e., within this window later scans were, if anything, more often APL). Engineered features kept their performance (CV AUROC 0.927 on all patients, 0.929 in the window), whereas the frozen CNN dropped from 0.957 to 0.858, stain and color features from 0.881 to 0.805, and engineered features computed on normalized images from 0.869 to 0.729. In every case the model score added information beyond scan time (likelihood-ratio $p < 0.001$), but scores and scan time were correlated (Spearman 0.23–0.55). Part of the high development performance of CNN and color features

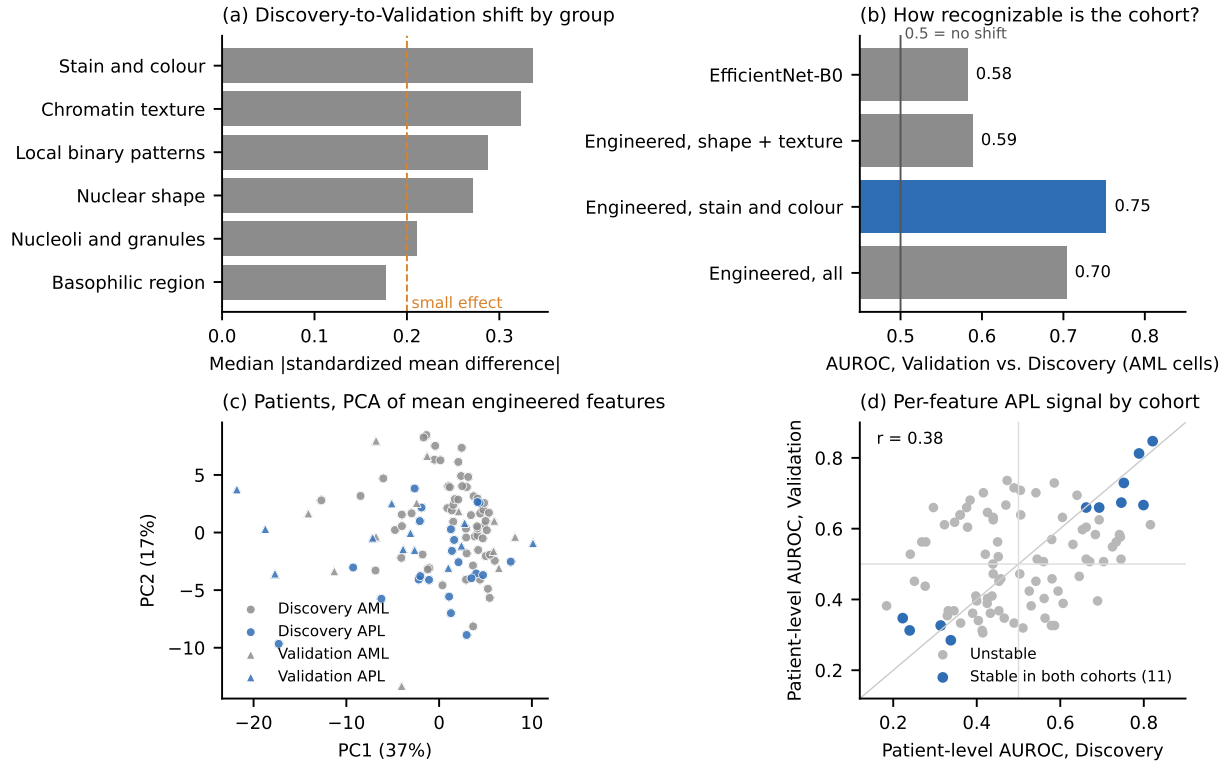


Fig. 4. Exploratory analysis of APL-AML. (a) Shift of engineered feature groups from Discovery to Validation. (b) AUROC of a classifier that tells the two cohorts apart from AML cells; 0.5 means no detectable shift. (c) Patients on the first two principal components of their mean engineered features. (d) Patient-level AUROC of each feature in the two cohorts; blue points separate the classes consistently.

TABLE IV

HOW MUCH OF THE GAP BETWEEN REPORTED AND REAL PERFORMANCE IS DUE TO LEAKAGE AND HOW MUCH TO THE EXTERNAL GENERALIZATION GAP (CELL LEVEL, MEAN OVER SEEDS; 95% CIS OF THE TWO COMPONENTS FROM 500 PATIENT-LEVEL BOOTSTRAP RESAMPLES: PAIRED OVER DEVELOPMENT PATIENTS FOR LEAKAGE, DEVELOPMENT AND EXTERNAL PATIENTS RESAMPLED INDEPENDENTLY FOR THE GAP). LEAKAGE = IMAGE-LEVEL CV AUROC − PATIENT-GROUPED CV AUROC (SAME DEVELOPMENT DATA, ONLY THE SPLIT DIFFERS). EXTERNAL GENERALIZATION GAP = PATIENT-GROUPED CV AUROC − EXTERNAL AUROC (NEW PATIENTS; TEMPORAL FOR APL-AML, A HELD-OUT SUBJECT FOLD FOR C-NMC); IT MIXES ALL DIFFERENCES BETWEEN COHORTS AND IS NOT A CAUSAL EFFECT OF ANY SINGLE ONE. LEAKY-TRAINED EXTERNAL: EXTERNAL AUROC OF THE MODELS TRAINED UNDER IMAGE-LEVEL FOLDS. RECALL, SPECIFICITY AND F1 ON THE EXTERNAL SET AT THE OPERATING POINT FIXED ON DEVELOPMENT DATA.

		AUROC				Decomposition of the gap		External, fixed operating point		
Task	Model	Image CV	Grouped CV	External [95% CI]	Leaky-trained ext.	Leakage	Ext. gap	Recall	Spec.	F1
T1 APL-AML: diagnosis on cells (patient label)	Engineered + LightGBM	0.987	0.896	0.669 [0.495, 0.811]	0.723	+0.091 [+0.045, +0.169]	+0.227 [+0.065, +0.432]	0.69	0.55	0.54
	Engineered + LR	0.938	0.895	0.658 [0.478, 0.798]	0.677	+0.042 [+0.019, +0.077]	+0.237 [+0.073, +0.421]	0.73	0.46	0.52
	EfficientNet-B0 + LR	0.975	0.867	0.694 [0.556, 0.811]	0.717	+0.108 [+0.072, +0.160]	+0.173 [+0.045, +0.323]	0.81	0.43	0.56
	Fusion + LR	0.985	0.901	**0.734 [0.595, 0.848]**	0.760	+0.083 [+0.050, +0.133]	+0.167 [+0.049, +0.311]	0.78	0.54	0.59
T2 APL-AML: blast detection (cell label)	Engineered + LightGBM	0.978	0.954	**0.957 [0.931, 0.977]**	0.956	+0.024 [+0.011, +0.040]	-0.003 [-0.032, +0.027]	0.99	0.62	0.58
	Engineered + LR	0.965	0.951	0.951 [0.922, 0.972]	0.951	+0.014 [+0.008, +0.021]	-0.000 [-0.031, +0.031]	0.98	0.70	0.63
	EfficientNet-B0 + LR	0.956	0.927	0.920 [0.874, 0.956]	0.920	+0.029 [+0.019, +0.041]	+0.007 [-0.034, +0.050]	0.96	0.64	0.58
	Fusion + LR	0.964	0.940	0.932 [0.894, 0.961]	0.932	+0.024 [+0.015, +0.037]	+0.009 [-0.025, +0.046]	0.96	0.68	0.61
T4 C-NMC: cells (subject label)	Engineered + LR	0.933	0.864	**0.840 [0.730, 0.902]**	0.848	+0.069 [+0.023, +0.143]	+0.023 [-0.089, +0.153]	0.95	0.58	0.89
	EfficientNet-B0 + LR	0.912	0.834	0.775 [0.640, 0.859]	0.786	+0.079 [+0.036, +0.149]	+0.059 [-0.053, +0.195]	0.96	0.43	0.87
	Fusion + LR	0.939	0.864	0.817 [0.692, 0.892]	0.825	+0.075 [+0.027, +0.149]	+0.047 [-0.070, +0.189]	0.95	0.54	0.88

therefore rests on time-linked acquisition differences, which would contribute to the external generalization gap when the model meets a later cohort; the 31-patient window is small, and these estimates are imprecise.

D. Choosing the Preprocessing

Table VII compares preprocessing options on a fixed subset of up to 60 cells per patient. Edge sharpening raised development performance (cells 0.873 to 0.887) but lowered external performance (cells 0.661 to 0.631; patients 0.667 to 0.639), widening the gap. CLAHE changed little (external 0.667 for cells and patients) and did not reduce the cohort shift (0.575 to 0.581). Reinhard normalization removed most of the shift (0.524) but lowered external cell AUROC (0.644). Macenko normalization reduced the shift (0.543) and was the only option that raised external performance at both levels (cells 0.690, patients 0.694), so it met the rule. Estimating stain vectors once per patient was faster but lowered patient-level development CV by 0.017 and failed the pre-specified rule, so normalization is done per cell (0.071 s per cell). On C-NMC, where no shift was detected, Macenko normalization lowered external AUROC slightly (subjects 0.882 to 0.857), supporting

TABLE V
THE SAME MODELS EVALUATED THE WAY MANY STUDIES REPORT THEM (IMAGE-LEVEL 5-FOLD CV, ACCURACY AT A 0.5 THRESHOLD) AND ON PATIENTS THEY HAVE NEVER SEEN (EXTERNAL SET). MEAN OVER SEEDS.

		Image-level CV (leaky)			External set (new patients)			
Task	Model	Accuracy	AUROC	F1	Accuracy	AUROC	Recall	F1
T1 APL-AML, diagnosis on cells	Engineered + LightGBM	0.950	0.987	0.942	0.691	0.669	0.245	0.342
	Fusion + LR	0.942	0.985	0.933	0.715	0.734	0.370	0.466
	Fine-tuned EfficientNet-B0	–	–	–	0.732	0.773	0.340	0.464
T2 APL-AML, blast detection	Engineered + LightGBM	0.928	0.978	0.875	0.898	0.957	0.895	0.786
	Fusion + LR	0.911	0.964	0.845	0.866	0.932	0.850	0.725
	Fine-tuned EfficientNet-B0	–	–	–	0.904	0.960	0.883	0.792
T4 C-NMC, cells	Engineered + LR	0.877	0.933	0.907	0.827	0.840	0.896	0.877
	Fusion + LR	0.894	0.939	0.921	0.816	0.817	0.895	0.871
	Fine-tuned EfficientNet-B0	–	–	–	0.825	0.841	0.946	0.882

TABLE VI
SCAN-TIME CONFOUNDING IN THE APL-AML DISCOVERY COHORT (82 PATIENTS). TIME PROXY: MEDIAN ANALYZER CELL NUMBER OF A PATIENT'S SIGNED CELLS. DIAGNOSIS MODELS: POOLED-STATISTICS LR, PATIENT-LEVEL 5-FOLD CV $\times$ 3 SEEDS, ON ALL PATIENTS AND ON THE PATIENTS SCANNED WHILE BOTH DIAGNOSES WERE BEING SCANNED (TIME WINDOW). SCORE ADDS TO TIME: LIKELIHOOD-RATIO TEST OF THE OUT-OF-FOLD SCORE ADDED TO A LOGISTIC MODEL ON THE TIME RANK. TIME FROM FEATURES: AUROC FOR EARLY VS. LATE SCAN AMONG AML PATIENTS ONLY.

Representation	CV AUROC, all	CV AUROC, time window	Score adds to time, p	Time from features (AML)
Engineered (all)	0.927	0.929	<0.001	0.794
Engineered: stain and colour	0.881	0.805	<0.001	0.760
Engineered: shape + texture	0.904	0.887	<0.001	0.767
EfficientNet-B0 (frozen)	0.957	0.858	<0.001	0.743
Engineered, on Macenko images	0.869	0.729	<0.001	–

Time proxy alone: Validation vs. Discovery 0.992; APL vs. AML in Discovery 0.775; within the window 0.325 (31 patients, 18 APL).

conditional use. These differences are small compared with the uncertainty of 24 external patients, and the subset included Validation patients, as disclosed above.

E. Engineered, Frozen and Fine-Tuned Features

For blast detection on new patients, engineered features with LightGBM reached an AUROC of 0.957 (95% CI 0.931–0.977) against 0.920 for frozen EfficientNet-B0 embeddings (difference 0.037; paired bootstrap 95% CI 0.008–0.065), and on C-NMC cells 0.840 against 0.775 (Table VIII). Fine-tuning closed this gap: external AUROC rose to 0.773 for T1 (frozen 0.694), 0.960 for blasts and 0.841 on C-NMC cells. For blasts it also gave a better operating point than the engineered features with LightGBM (specificity 0.83 instead of 0.62 at a recall of 0.95–0.99, F1 0.74 instead of 0.58) and than the frozen CNN (specificity 0.64, F1 0.58). In an ablation, removing the stain and color group cost blast detection the most (-0.034 AUROC, 95% CI -0.060 to -0.015; Table S15, Fig. S11). For T1, the threshold chosen on Discovery flagged almost every Validation cell (recall 0.996, specificity 0.17), the operating-point problem seen throughout the temporal cohort.

F. Dimensionality Reduction: Before and After PCA

Table IX and Fig. 5 show the effect of PCA on the pooled patient model (natural scale; full sweep in Table S6). Without reduction, each patient is described by 6,400 values, about 100 per APL-AML training patient.

On APL-AML without stain normalization, PCA raised external AUROC from 0.708 to 0.792 with 64 components but lowered the decisions at the clinical threshold (6 instead of 8 of 12 APL patients detected; F1 0.57 instead of 0.67). On C-NMC it cost a little on every metric (AUROC 0.908 without, 0.853 with 64 components; 16 instead of 17 ALL subjects detected). In pooled CV over all 106 APL-AML patients, 32 and 64 components ranked as well as the unreduced embedding (ΔAUROC $+0.000$ and $+0.006$; Table S9), and 16 components were worse (-0.057). The pre-declared rule (highest development CV among $k \in \{16, 32, 64\}$) selected 64 on both datasets.

PCA and stain normalization interacted. After Macenko normalization, PCA-64 no longer improved ranking: external AUROC fell from 0.701 to 0.639 and F1 from 0.79 to 0.76, with recall unchanged (11 of 12). Normalization had already removed part of the cohort-specific color variation that, without it, PCA removed together with some signal. PCA keeps directions of large variance, not those that separate the classes, and the external disease signal grew steadily with the number of components on both datasets (Fig. 5c). We therefore use PCA only to reduce complexity (Section IV-H), not as a way to gain accuracy.

Whitening was a pitfall: with each component standardized, development CV rose to 0.99 while external AUROC on C-NMC fell from 0.849 (8 components) to 0.655 (256), the signature of a classifier fitting noise directions (dashed lines in Fig. 5).

G. Training Settings: Epochs and Early Stopping

We compared the fixed 40 epochs of attention MIL with a data-driven alternative: inside every training fold, nested 3-fold patient-level CV chose the epoch with the lowest validation loss. This did not help (Table S7). External AUROC changed by less than 0.01 on APL-AML and fell by 0.02–0.06 on C-NMC, and F1 and calibration did not improve. Because the 80/20 calibration split is taken first, each inner model was trained on about 35 and validated on about 17 APL-AML

TABLE VII
PREPROCESSING BEFORE FEATURE EXTRACTION (FROZEN EFFICIENTNET-B0, UP TO 60 CELLS PER PATIENT, SAME FOLDS AND CLASSIFIERS FOR EVERY VARIANT). COHORT SHIFT: AUROC FOR ASSIGNING CONTROL-CLASS CELLS TO THEIR COHORT (0.5 = NO DETECTABLE SHIFT). GAP: DEVELOPMENT CV MINUS EXTERNAL AUROC. THE RULE FIXED IN ADVANCE SELECTED THE METHOD THAT REDUCED THE SHIFT WITHOUT LOWERING EXTERNAL PERFORMANCE. C-NMC, WHICH SHOWED NO SHIFT, IS GIVEN FOR COMPARISON.

		Cells (T1)			Patients (T3)		
Preprocessing	Cohort shift	Dev. CV (SD; worst fold)	External	Gap	Dev. CV (SD; worst fold)	External	Gap
APL-AML (6,360 cells, 106 patients)							
None	0.575	0.873 (0.036; 0.818)	0.661	0.212	0.956 (0.047; 0.875)	0.667	0.289
Edge sharpening (unsharp mask)	0.561	0.887 (0.030; 0.834)	0.631	0.256	0.963 (0.050; 0.875)	0.639	0.324
Contrast enhancement (CLAHE)	0.581	0.861 (0.024; 0.827)	0.667	0.194	0.939 (0.057; 0.833)	0.667	0.273
Reinhard colour normalization	0.524	0.873 (0.026; 0.836)	0.644	0.228	0.972 (0.039; 0.896)	0.681	0.291
Macenko stain normalization (selected)	0.543	0.850 (0.037; 0.785)	0.690	0.160	0.942 (0.052; 0.875)	0.694	0.247
Macenko, stain vectors per patient	0.554	0.859 (0.048; 0.775)	0.615	0.244	0.925 (0.070; 0.833)	0.611	0.314
C-NMC (3,773 cells, 73 subjects; no shift detected)							
None	0.549	0.832 (0.050; 0.753)	0.746	0.086	0.889 (0.222; 0.444)	0.882	0.007
Macenko stain normalization	0.560	0.818 (0.058; 0.708)	0.729	0.088	0.861 (0.238; 0.389)	0.857	0.004

TABLE VIII
ENGINEERED, FROZEN AND FINE-TUNED FEATURES AT CELL LEVEL ON BOTH DATASETS. THE FINE-TUNED MODEL ADAPTS THE LAST TWO STAGES OF EFFICIENTNET-B0 (TABLE S1; ONE SEED); THE OTHER MODELS ARE AVERAGED OVER THREE SEEDS. LEAKAGE: IMAGE-LEVEL MINUS PATIENT-GROUPED CV AUROC; EXTERNAL GAP: GROUPED CV MINUS EXTERNAL AUROC. RECALL, SPECIFICITY AND F1 ON THE EXTERNAL SET AT THE OPERATING POINT FIXED ON DEVELOPMENT DATA. BEST EXTERNAL AUROC PER TASK IN BOLD.

Task	Model	Image CV	Grouped CV	External [95% CI]	Leakage	Ext. gap	Recall	Spec.	F1
T1 APL-AML, diagnosis on cells	Engineered + LightGBM	0.987	0.896	0.669 [0.495, 0.811]	+0.091	+0.227	0.69	0.55	0.54
	Engineered + LR	0.938	0.895	0.658 [0.478, 0.798]	+0.042	+0.237	0.73	0.46	0.52
	EfficientNet-B0 + LR	0.975	0.867	0.694 [0.556, 0.811]	+0.108	+0.173	0.81	0.43	0.56
	Fusion + LR	0.985	0.901	0.734 [0.595, 0.848]	+0.083	+0.167	0.78	0.54	0.59
	Fine-tuned EfficientNet-B0	–	0.875	**0.773 [0.612, 0.890]**	–	+0.101	1.00	0.17	0.55
T2 APL-AML, blast detection	Engineered + LightGBM	0.978	0.954	0.957 [0.931, 0.977]	+0.024	-0.003	0.99	0.62	0.58
	Engineered + LR	0.965	0.951	0.951 [0.922, 0.972]	+0.014	-0.000	0.98	0.70	0.63
	EfficientNet-B0 + LR	0.956	0.927	0.920 [0.874, 0.956]	+0.029	+0.007	0.96	0.64	0.58
	Fusion + LR	0.964	0.940	0.932 [0.894, 0.961]	+0.024	+0.009	0.96	0.68	0.61
	Fine-tuned EfficientNet-B0	–	0.968	**0.960 [0.928, 0.981]**	–	+0.008	0.95	0.83	0.74
T4 C-NMC, cells	Engineered + LightGBM	0.972	0.876	0.839 [0.694, 0.921]	+0.096	+0.037	0.91	0.61	0.88
	Engineered + LR	0.933	0.864	0.840 [0.730, 0.902]	+0.069	+0.023	0.95	0.58	0.89
	EfficientNet-B0 + LR	0.912	0.834	0.775 [0.640, 0.859]	+0.079	+0.059	0.96	0.43	0.87
	Fusion + LR	0.939	0.864	0.817 [0.692, 0.892]	+0.075	+0.047	0.95	0.54	0.88
	Fine-tuned EfficientNet-B0	–	0.853	**0.841 [0.715, 0.917]**	–	+0.012	0.99	0.42	0.88

TABLE IX
IMPACT OF PCA ON PATIENT/SUBJECT-LEVEL DETECTION (EXTERNAL SETS). BEFORE: UNREDUCED 1,280-DIMENSIONAL EMBEDDING. AFTER: PCA FITTED ON DEVELOPMENT CELLS, SCORES KEPT AT THEIR NATURAL SCALE. RULE: NUMBER OF COMPONENTS WITH THE HIGHEST DEVELOPMENT CV AUROC WITHIN THE PRE-SPECIFIED SET $\{16, 32, 64\}$; OVER THE FULL SWEEP (4–256 AND NO PCA), DEVELOPMENT CV FAVOURED 256 COMPONENTS ON APL-AML AND NO REDUCTION ON C-NMC. THRESHOLD METRICS AT THE OPERATING POINT FIXED ON DEVELOPMENT DATA.

Dataset	Representation	AUROC	ΔAUROC	Acc.	Recall (missed)	Spec.	F1	NPV
APL-AML	Before: no PCA	0.708	–	0.67	0.67 (4/12)	0.67	0.67	0.67
	After: PCA-32	0.785	+0.076	0.62	0.50 (6/12)	0.75	0.57	0.60
	After: PCA-64 (rule)	0.792	+0.083	0.62	0.50 (6/12)	0.75	0.57	0.60
	PCA-256 (best dev. CV, full sweep)	0.778	+0.069	0.62	0.50 (6/12)	0.75	0.57	0.60
C-NMC	Before: no PCA	0.908	–	0.84	1.00 (0/17)	0.64	0.87	1.00
	After: PCA-32	0.840	-0.067	0.71	0.94 (1/17)	0.43	0.78	0.86
	After: PCA-64 (rule)	0.853	-0.055	0.74	0.94 (1/17)	0.50	0.80	0.88

patients (C-NMC: 18 and 9), and the chosen epoch ranged from 1 to 40 for the same model, so early stopping made results less stable across seeds. We kept the fixed schedule. Learning curves (Fig. S7, computed with whitened PCA inputs from our first analysis and after the configuration was fixed) show that held-out AUROC plateaued after about 10 epochs while held-out loss rose, i.e., the fixed schedule cost no ranking but produced overconfident probabilities. Replacing average pooling of the fine-tuned CNN by attention pooling over spatial positions changed development AUROC by -0.020, $+0.014$ and $+0.005$ on three folds and was not pursued.

H. Final Pipelines on New Patients

Table X and Fig. 6 trace how each component changed performance on new patients (all six benchmark patient-level models and their confusion matrices are in Table S8 and Fig. S8); Table XI sets the pipelines against their cost.

Primary pipeline. On APL-AML, the primary pipeline detected all 12 APL patients in the Validation cohort (recall 1.00, exact 95% CI 0.74–1.00; AUROC 0.806, 0.615–0.956), but flagged 10 of 12 AML patients (specificity 0.17, 0.02–0.48; F1 0.71). On C-NMC (no normalization, because no shift was detected) it reached an AUROC of 0.960 and detected 16 of 17 ALL subjects (recall 0.94, 0.71–1.00) at a specificity of 0.79 (0.49–0.95; F1 0.89). With 12 APL patients, a perfect

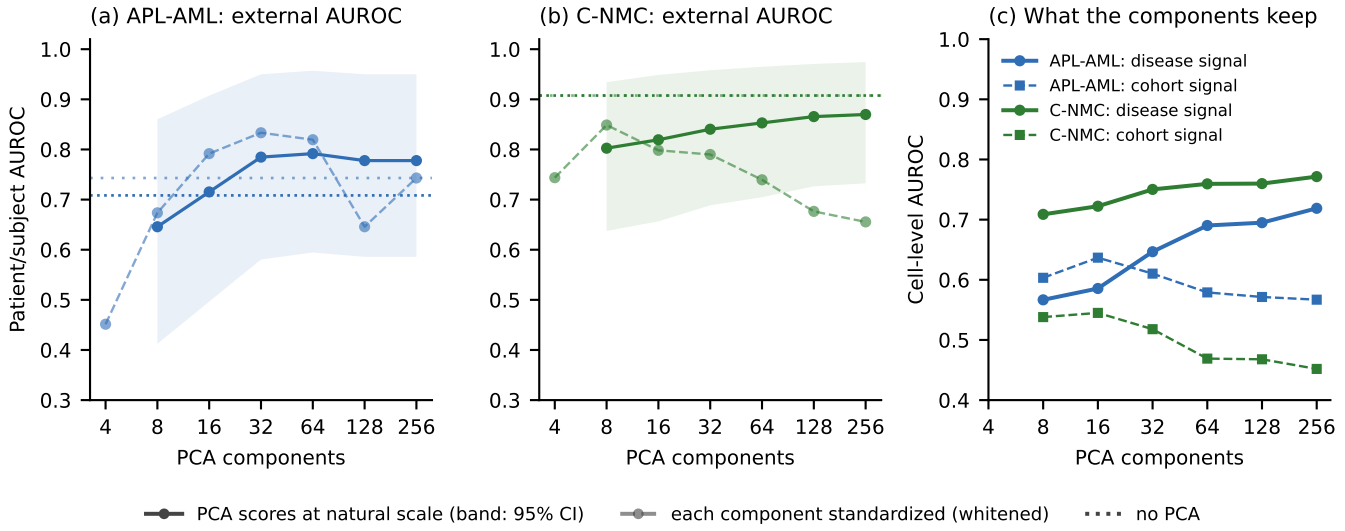


Fig. 5. PCA of EfficientNet-B0 embeddings, pooled patient/subject model. (a) APL-AML and (b) C-NMC: external AUROC versus the number of components at natural scale (solid; band: 95% CI) and whitened (dashed); horizontal lines: no PCA. (c) External cell-level disease signal (circles) and cohort signal for control-class cells (squares; 0.5 = none).

observed recall is compatible with a true recall as low as 0.74. The ensemble did not rank better than its best member (APL-AML: attention MIL 0.812; C-NMC: fine-tuned CNN 0.962), but it gave the most balanced operating point: on APL-AML it matched the pooled engineered model (no missed patient, specificity 0.17) where attention MIL flagged every AML patient and the fine-tuned CNN missed two APL patients; on C-NMC it kept the specificity of the pooled model (0.79) where the fine-tuned CNN reached 0.57. Its value is robustness of the operating point, not better ranking.

What each step contributed. Fine-tuning was the most consistent improvement at cell level on both datasets. Stain normalization helped the frozen CNN at the operating point (1 instead of 4 missed APL patients; F1 0.79 instead of 0.67) but not the fine-tuned CNN (AUROC 0.771 vs. 0.778), which appears to absorb much of the color shift itself.

Sensitivity to stain normalization. Because Macenko normalization was selected on a subset that included Validation patients and did not reduce CNN-detectable shift on the full data, we re-ran the primary pipeline without it. External AUROC fell from 0.806 to 0.785, with the same operating point (12 of 12 APL patients detected, specificity 0.17), so the reported result does not depend on normalization. Computing the engineered features on normalized images as well, instead of on the originals, lowered external AUROC to 0.663 (engineered pooled LR alone: 0.757 to 0.576) and missed one APL patient: normalization removes the color information that makes engineered features diagnostic, which supports applying it to the CNN branch only (Table X).

Secondary pipeline and the cost of PCA. With PCA-64, the patient-level classifier has 321 instead of 6,401 parameters and stores 256 instead of 5,120 bytes per cell; the classifier itself fits in half the time (0.13 vs. 0.26 s), but fitting PCA adds about 6 s per training set, so training as a whole is slower; attention MIL on the reduced inputs would have 7 times fewer parameters. The price on APL-AML was 0.06 AUROC (0.701 to 0.639), 0.03 F1 (0.79 to 0.76) and 0.08 specificity, with the same recall. Image-level steps dominate the time per patient—Macenko normalization 0.071 s, the CNN 0.051 s and engineered features 0.038 s per cell—so the primary pipeline needs about 80 s per 500-cell patient on our CPU, 45% of it for normalization, and PCA mainly saves storage, memory and model size rather than time.

I. Replication on C-NMC

Table XII repeats the experiments on C-NMC. All six patient-level models ranked the 31 external subjects well (AUROC 0.853–0.941), engineered features best (pooled LR 0.941). Unlike on APL-AML, the development operating point transferred: every model detected at least 15 of the 17 ALL subjects, at specificities up to 0.79. C-NMC external ALL subjects contributed more cells than normal subjects (median 161 vs. 41; count alone AUROC 0.735), whereas in the development folds the count carried no signal (0.453). The model scores were nearly uncorrelated with the count ($|\rho| \leq 0.29$), remained discriminative after regressing it out (AUROC 0.89–0.95), and added information beyond it ($p < 0.001$; Table S12), so the number of cells does not explain the C-NMC results.

J. A Hematology Foundation Model

Table XIII compares frozen DinoBloom-S features with the other representations. On C-NMC cells, DinoBloom outperformed the frozen ImageNet network on development (0.859 vs. 0.834) and external data (0.801 vs. 0.775), and combined with engineered features it had the highest development CV of all cell-level models (0.882), which also gave the highest external AUROC (0.845) and F1 score (0.89). At subject level, the same combination with pooled LR again ranked

TABLE X

WHAT EACH ADDITION CHANGES ON NEW PATIENTS (EXTERNAL SETS; CELLS: T1/T4, PATIENTS: T3/T5). EACH BLOCK STARTS FROM A FROZEN IMAGENET CNN; Δ: AUROC CHANGE RELATIVE TO THE FIRST ROW OF THE BLOCK. THRESHOLD METRICS AT THE OPERATING POINT FIXED ON DEVELOPMENT DATA (TARGET RECALL 0.95); PATIENT ROWS GIVE EXACT 95% CIS FOR RECALL AND SPECIFICITY. CELL ROWS: MEAN OF THREE SEEDS (FROZEN) OR ONE SEED (FINE-TUNED). THE ENSEMBLE AVERAGES PERCENTILE SCORES OF THE FINE-TUNED CNN (WITH MACENKO ON APL-AML), ENGINEERED POOLED LR AND ENGINEERED ATTENTION MIL.

Dataset	Level	Model	AUROC [95% CI]	ΔAUROC	Accuracy	Recall	Missed	Specificity	F1
APL-AML	Cell	Frozen CNN + LR	0.694 [0.556, 0.811]	–	0.56	0.81	–	0.43	0.56
		Frozen CNN + engineered (fusion)	0.734 [0.595, 0.848]	+0.040	0.62	0.78	–	0.54	0.59
		+ fine-tuning	0.773 [0.612, 0.890]	+0.079	0.45	1.00	–	0.17	0.55
		+ fine-tuning + Macenko	0.752 [0.601, 0.864]	+0.058	0.44	0.99	–	0.15	0.55
APL-AML	Patient	Frozen CNN, pooled + LR	0.708 [0.493, 0.907]	–	0.67	0.67 (0.35–0.90)	4/12	0.67 (0.35–0.90)	0.67
		+ PCA-64	0.792 [0.594, 0.957]	+0.083	0.62	0.50 (0.21–0.79)	6/12	0.75 (0.43–0.95)	0.57
		Frozen CNN + Macenko	0.701 [0.482, 0.907]	-0.007	0.75	0.92 (0.62–1.00)	1/12	0.58 (0.28–0.85)	0.79
		Frozen CNN + Macenko + PCA-64 (secondary)	0.639 [0.400, 0.867]	-0.069	0.71	0.92 (0.62–1.00)	1/12	0.50 (0.21–0.79)	0.76
		Fine-tuned CNN, mean of cells	0.778 [0.576, 0.938]	+0.069	0.75	0.83 (0.52–0.98)	2/12	0.67 (0.35–0.90)	0.77
		Fine-tuned CNN + Macenko	0.771 [0.563, 0.941]	+0.062	0.67	0.83 (0.52–0.98)	2/12	0.50 (0.21–0.79)	0.71
		Engineered, pooled + LR	0.757 [0.555, 0.938]	+0.049	0.58	1.00 (0.74–1.00)	0/12	0.17 (0.02–0.48)	0.71
		Engineered, attention MIL	0.812 [0.614, 0.963]	+0.104	0.50	1.00 (0.74–1.00)	0/12	0.00 (0.00–0.26)	0.67
		Primary: ensemble (Macenko fine-tuned + engineered)	0.806 [0.615, 0.956]	+0.097	0.58	1.00 (0.74–1.00)	0/12	0.17 (0.02–0.48)	0.71
		Sensitivity: primary without Macenko	0.785 [0.585, 0.944]	+0.076	0.58	1.00 (0.74–1.00)	0/12	0.17 (0.02–0.48)	0.71
		Sensitivity: primary, engineered also on Macenko images	0.663 [0.427, 0.888]	-0.045	0.58	0.92 (0.62–1.00)	1/12	0.25 (0.05–0.57)	0.69
C-NMC	Cell	Frozen CNN + LR	0.775 [0.640, 0.859]	–	0.79	0.96	–	0.43	0.87
		Frozen CNN + engineered (fusion)	0.817 [0.692, 0.892]	+0.042	0.83	0.95	–	0.54	0.88
		+ fine-tuning	0.841 [0.715, 0.917]	+0.067	0.81	0.99	–	0.42	0.88
C-NMC	Patient	Frozen CNN, pooled + LR	0.908 [0.786, 0.992]	–	0.84	1.00 (0.80–1.00)	0/17	0.64 (0.35–0.87)	0.87
		+ PCA-64	0.853 [0.705, 0.965]	-0.055	0.74	0.94 (0.71–1.00)	1/17	0.50 (0.23–0.77)	0.80
		Fine-tuned CNN, mean of cells	0.962 [0.887, 1.000]	+0.055	0.81	1.00 (0.80–1.00)	0/17	0.57 (0.29–0.82)	0.85
		Engineered, pooled + LR	0.941 [0.840, 1.000]	+0.034	0.87	0.94 (0.71–1.00)	1/17	0.79 (0.49–0.95)	0.89
		Engineered, attention MIL	0.912 [0.791, 1.000]	+0.004	0.77	0.88 (0.64–0.99)	2/17	0.64 (0.35–0.87)	0.81
		Primary: ensemble of the three members above	0.960 [0.886, 1.000]	+0.053	0.87	0.94 (0.71–1.00)	1/17	0.79 (0.49–0.95)	0.89

TABLE XI

PERFORMANCE AND COMPLEXITY ON APL-AML (24 EXTERNAL PATIENTS). TIME PER PATIENT ASSUMES 500 CELLS ON OUR CPU (2 VCPUS): MACENKO NORMALIZATION 0.071 S, CNN FORWARD PASS 0.051 S AND ENGINEERED FEATURES 0.038 S PER CELL; CLASSIFIER TIMES ARE MEASURED. STORAGE: BYTES KEPT PER CELL AFTER FEATURE EXTRACTION. PARAMETERS AND TRAINING TIME REFER TO THE PATIENT-LEVEL CLASSIFIER (INCLUDING PCA FITTING).

	Performance (external, fixed threshold)							Complexity			
Pipeline	AUROC	Acc.	Recall	Missed	Spec.	F1	NPV	Storage/cell	Parameters	Training (s)	Time/patient (s)
Primary: Macenko + fine-tuned CNN + engineered, ensemble	0.806	0.58	1.00 (0.74–1.00)	0/12	0.17 (0.02–0.48)	0.71	1.00	5,492 B	4,038,033	631.00	80
Macenko + frozen CNN, no PCA	0.701	0.75	0.92 (0.62–1.00)	1/12	0.58 (0.28–0.85)	0.79	0.88	5,120 B	6,401	0.26	61
Secondary: Macenko + frozen CNN + PCA-64	0.639	0.71	0.92 (0.62–1.00)	1/12	0.50 (0.21–0.79)	0.76	0.86	256 B	321	6.15	61

TABLE XII

REPLICATION ON THE SECOND DATASET, C-NMC 2019 (ALL VS. NORMAL B CELLS; DEVELOPMENT: OFFICIAL FOLDS 0–1, 42 SUBJECTS; EXTERNAL: FOLD 2, 31 UNSEEN SUBJECTS, 17 WITH ALL). SAME FEATURES, MODELS, PROTOCOL AND OPERATING-POINT RULE AS FOR APL-AML. LEAKAGE: IMAGE-LEVEL MINUS SUBJECT-GROUPED CV AUROC; EXTERNAL GAP: GROUPED CV MINUS EXTERNAL AUROC.

(a) Cell level: every cell carries its subject's diagnosis (mean of 3 seeds)

Model	Image CV	Grouped CV	External [95% CI]	Recall	Spec.	F1	Leakage	Ext. gap
Engineered + LightGBM	0.972	0.876	0.839 [0.694, 0.921]	0.91	0.61	0.88	+0.096	+0.037
Engineered + LR	0.933	0.864	0.840 [0.730, 0.902]	0.95	0.58	0.89	+0.069	+0.023
EfficientNet-B0 + LR	0.912	0.834	0.775 [0.640, 0.859]	0.96	0.43	0.87	+0.079	+0.059
Fusion + LR	0.939	0.864	0.817 [0.692, 0.892]	0.95	0.54	0.88	+0.075	+0.047

(b) Subject level: is this subject's sample leukemic?

Representation	Aggregation	CV AUROC	External [95% CI]	Recall (95% CI) [missed]	Spec. (95% CI)	F1	NPV
Engineered	Pooled + LR	0.906	0.941 [0.840, 1.000]	0.94 (0.71–1.00) [1]	0.79 (0.49–0.95)	0.89	0.92
EfficientNet-B0	Pooled + LR	0.855	0.920 [0.807, 0.992]	0.94 (0.71–1.00) [1]	0.71 (0.42–0.92)	0.86	0.91
Fusion	Pooled + LR	0.878	0.929 [0.824, 0.996]	1.00 (0.80–1.00) [0]	0.71 (0.42–0.92)	0.89	1.00
Engineered	Attention MIL	0.881	0.912 [0.791, 1.000]	0.88 (0.64–0.99) [2]	0.64 (0.35–0.87)	0.81	0.82
EfficientNet-B0	Attention MIL	0.874	0.853 [0.704, 0.963]	1.00 (0.80–1.00) [0]	0.43 (0.18–0.71)	0.81	1.00
Fusion	Attention MIL	0.887	0.853 [0.696, 0.967]	1.00 (0.80–1.00) [0]	0.57 (0.29–0.82)	0.85	1.00

best on development data (0.919) and detected all 17 leukemic subjects in the external fold (AUROC 0.962, specificity 0.71). On the later APL-AML cohort, DinoBloom with attention MIL had the highest development CV (0.924) and the highest external AUROC of any model in this study (0.868, 95% CI 0.70–0.99), but at the development threshold it detected only 6 of 12 APL patients; combined with engineered features it missed one at a specificity of 0.58. Domain-specific pretraining thus improved ranking on all three test sets without fixing the operating point.

Before turning to DinoBloom we tried to improve the fine-tuned CNN on C-NMC, selecting on development CV

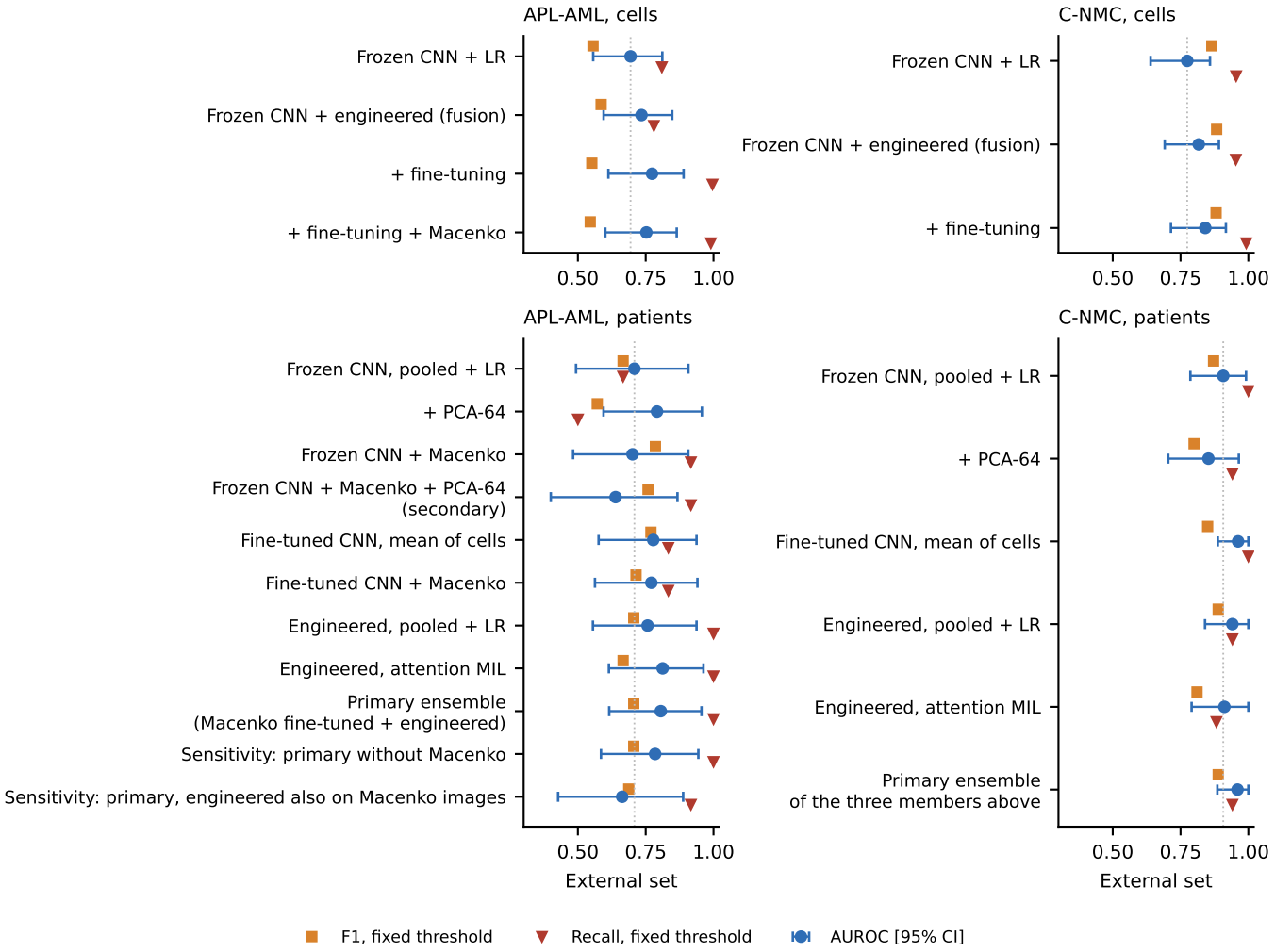


Fig. 6. Effect of each added component on new patients (external sets): AUROC with 95% CI (circles), and F1 score (squares) and recall (triangles) at the operating point fixed on development data. Dotted line: AUROC of the first row.

only. Cropping each image to the cell raised cell-level development AUROC from 0.853 to 0.860 but lowered subject-level development AUROC (0.864 to 0.833), well within the variation between folds (0.64–0.94); fine-tuning one more network stage overfit within six epochs (training loss near zero, held-out AUROC 0.82–0.83 on the first two folds) and was stopped; and LightGBM on engineered features gained 0.012 in development CV but nothing on the external fold (0.839 vs. 0.840). A SHAP analysis of the engineered-feature classifier (Fig. S10) attributed malignancy mostly to nuclear color, larger nuclear and cell size and local texture patterns, consistent with lymphoblast morphology, although the weight on color may partly reflect subject-specific staining.

TABLE XIII
HEMATOLOGY FOUNDATION MODEL (DINOBLOOM-S, FROZEN, CLS TOKEN) COMPARED WITH IMAGENET, FINE-TUNED AND ENGINEERED FEATURES. SELECTION BY DEVELOPMENT CV ONLY (BOLD: HIGHEST DEVELOPMENT CV IN EACH BLOCK). EXTERNAL: C-NMC FOLD 2 AND THE APL-AML VALIDATION COHORT; RECALL AND SPECIFICITY AT THE DEVELOPMENT THRESHOLD.

Block	Features	Dev. CV	External [95% CI]	Recall	Spec.	F1
C-NMC, cells	EfficientNet-B0 (ImageNet), frozen	0.834	0.775 [0.640, 0.859]	0.96	0.43	0.87
	DinoBloom-S, frozen	0.859	0.801 [0.661, 0.889]	0.94	0.47	0.86
	EfficientNet-B0, fine-tuned	0.853	0.841 [0.715, 0.917]	0.99	0.42	0.88
	Engineered	0.864	0.840 [0.730, 0.902]	0.95	0.58	0.89
	Engineered + LightGBM	0.876	0.839 [0.694, 0.921]	0.91	0.61	0.88
	EfficientNet-B0 + engineered	0.864	0.817 [0.692, 0.892]	0.95	0.54	0.88
	DinoBloom-S + engineered	0.882	0.845 [0.724, 0.917]	0.94	0.59	0.89
C-NMC, subjects	Engineered, pooled LR	0.906	0.941 [0.840, 1.000]	0.94	0.79	0.89
	DinoBloom-S, pooled LR	0.906	0.941 [0.849, 1.000]	0.94	0.64	0.84
	DinoBloom-S, attention MIL	0.913	0.945 [0.853, 1.000]	0.88	0.79	0.86
	DinoBloom-S + engineered, pooled LR	0.919	0.962 [0.891, 1.000]	1.00	0.71	0.89
APL-AML, later cohort	Engineered, attention MIL	0.827	0.812 [0.614, 0.963]	1.00	0.00	0.67
	DinoBloom-S, pooled LR	0.880	0.840 [0.652, 0.979]	0.83	0.67	0.77
	DinoBloom-S, attention MIL	0.924	0.868 [0.703, 0.993]	0.50	0.92	0.63
	DinoBloom-S + engineered, pooled LR	0.869	0.833 [0.644, 0.979]	0.92	0.58	0.79

K. Independent External Test

The independent cohort differed strongly from Discovery: AML cells of the two sources could be told apart almost perfectly (AUROC 0.999 on stain and color features, permutation $p \leq 0.03$), so by the pre-specified rule Macenko normalization was applied to the CNN branch. Table XIV and Fig. 8 report the single prespecified analysis. The primary pipeline, trained on 82 Discovery patients from another hospital, ranked the 129 independent patients with an AUROC of 0.862 (95% CI 0.765–0.936), higher than on the later APL-AML cohort, and the pre-declared DinoBloom combination reached 0.877

(0.773–0.960). Engineered features carried most of the transfer (pooled LR 0.832), whereas the ImageNet networks transferred worst despite stain normalization (fine-tuned 0.737, secondary pipeline with PCA 0.660). The development threshold, however, failed to maintain an acceptable operating point: the primary pipeline detected all 24 APL patients (recall 1.00, 95% CI 0.86–1.00) but cleared none of the 105 AML patients, and the other models behaved similarly (specificity 0.00–0.18), except the secondary pipeline, which missed 6 APL patients.

Can the ranking reduce workload? Because no fixed threshold transferred, we evaluated the potential workload reduction from ranking alone, for a laboratory that refers patients for urgent review in order of their score (Table XV, Fig. 7; post hoc, no model changed). Referring the top 20% of independent-cohort patients captured 15 of 24 APL patients (62.5%) with the primary pipeline and 17 of 24 (70.8%) with the DinoBloom combination (average precision 0.58 and 0.74 against a base rate of 0.19), but capturing 95% of APL patients required reviewing 60% and 75% of all patients. The ranking concentrates APL at the top, but at a high-sensitivity target it spares only a quarter to two fifths of the review workload; it does not by itself establish an efficient triage system. Whether a reduction of this size is worthwhile depends on the comparator workflow, the review capacity and the cost of a delayed APL diagnosis, none of which we measured; the plausible role supported by these data is retrospective, ranking-based prioritization that warrants prospective testing.

Local threshold adaptation, validated on separate patients. We split the independent cohort at random into a local calibration sample and an evaluation sample (2,000 splits; Table S13). A threshold just below the lowest calibration APL score reached, on the evaluation patients, the mean recall expected from exchangeability, $m/(m+1)$: 0.84 with $m = 5$ and 0.93 with $m = 12$ local APL cases (Fig. 8b). Single laboratories, however, would see much less: in the worst 5% of splits recall was 0.53 with $m = 5$ and 0.75 with $m = 12$, and at $m = 12$ the primary pipeline referred 70% of patients. Using local AML cases as well (12 APL and 50 AML) and maximizing calibration specificity at a calibration recall of at least 0.9 doubled specificity (0.35 to 0.69) at the cost of recall (0.93 to 0.86). The expected-recall rule is a property of threshold setting, not evidence about the resulting rule: validating it is a separate step on separate cases. Under an independent binomial sampling assumption, detecting all of 59 APL patients in a separate validation set at the locked threshold gives a one-sided exact 95% lower confidence bound for sensitivity of about 0.95 (29 patients for 0.90); this concerns sensitivity only and says nothing about specificity, workload or robustness to further shift.

Reverse transfer (exploratory). Trained on the 129 MLL patients, with settings and thresholds from patient-grouped CV within MLL, the same model families ranked the 106 APL-AML patients with AUROCs of 0.75–0.78 (Table XVI), lower than in the forward direction (0.83–0.88), and lower in the later APL-AML period than in the earlier one (DinoBloom combination 0.69 vs. 0.79). Attention MIL did not transfer (0.54). The thresholds failed in the opposite direction: the MLL-trained models missed most APL patients (recall 0.00–0.47) at high specificity (0.83–1.00). Transfer was thus not symmetric, and in both directions discrimination degraded much less than the operating point. Because the APL-AML cohort had been examined extensively during development, this direction is exploratory.

L. Calibration

Patient-level probabilities were poorly calibrated on both external sets (Table S11, Fig. S9). On APL-AML, calibration intercepts were positive (1.0–1.3 for raw outputs), i.e., APL risk was underestimated, as expected when the APL prevalence rises from 27% to 50%; on C-NMC they were negative (-0.7 to -1.6) as the ALL prevalence fell from 71% to 55%. Slopes varied widely (0.3–5.4), and the Brier score of the primary pipeline after Platt scaling on development data was 0.231 on APL-AML and 0.136 on C-NMC. With 24 and 31 external patients, these estimates are imprecise, but they show that predicted probabilities should not be used as risks without local recalibration.

M. Pooled Analysis and Local Threshold Adaptation

In pooled CV over all 106 APL-AML patients (Table S9), the unreduced EfficientNet-B0 embedding ranked patients best (AUROC 0.953, with per-feature standardization; 0.929 at natural scale), better than engineered features with pooled statistics (-0.059, $p = 0.007$ unadjusted, $p = 0.052$ after Holm correction) and the promyelocyte fraction used as a manual biomarker (0.677). Thresholds chosen inside each training fold transferred to the held-out folds (recall 0.89–1.00), which supports the view that the failure on the Validation cohort reflects temporal shift; because patients from both periods were mixed, this analysis does not measure transfer to a later cohort.

Setting a new site's threshold just below the lowest score of $m = k/2$ local APL cases gave the same expected recall for every model, about $m/(m+1)$ (Fig. 9a; Table S10). This follows from exchangeability and does not depend on the model: about 19 confirmed local APL cases are needed before such a threshold can be *expected* to reach a recall of 0.95, which is an average over laboratories, not a guarantee for any one of them (Section IV-K). Lowering the development threshold only when a local APL case would otherwise be missed changed little for models whose development threshold already reached a recall of 0.92–1.00 on the Validation cohort; it mainly cost specificity (frozen CNN with Macenko: 0.58 to 0.42 with eight local cases), and raised the recall of the fine-tuned CNN only from 0.83 to 0.86 (Fig. 9b). For the models with recall below 1.00, the probability of missing at least one APL patient among the remaining 16 stayed at 0.66–0.68. At an APL prevalence of 10% (Fig. 9c), the primary pipeline gave the highest net benefit at a 10% threshold probability (0.050 against 0 for testing everyone and 0.017–0.042 for the other models), and benefits close to testing everyone below 5%.

N. Comparison With Existing Methods on the Same Cohort

Table XVII compares our pipelines with existing methods on the same patients. Sidhom *et al.* reported an AUC of 0.743

TABLE XIV

INDEPENDENT EXTERNAL TEST (PRESPECIFIED ANALYSIS PLAN, COMMIT ACF173B): MODELS TRAINED ON THE APL-AML DISCOVERY COHORT ONLY AND APPLIED ONCE TO THE AML-CYTOMORPHOLOGY_MLL_HELMHOLTZ COHORT (ANOTHER LABORATORY, COUNTRY, SCANNER AND STAIN; 24 APL AND 105 OTHER AML PATIENTS, UP TO 200 CELLS EACH). LABEL-FREE SHIFT TEST: AUROC 0.999 FOR TELLING MLL FROM DISCOVERY AML CELLS BY STAIN AND COLOR ($p \leq 0.03$), SO MACENKO NORMALIZATION WAS APPLIED TO THE CNN BRANCH. THRESHOLD METRICS AT THE DISCOVERY OPERATING POINT (RECALL $\geq$0.95), WITH EXACT 95% CIS; NPV N/A: NO PATIENT CALLED NEGATIVE.

Model	AUROC [95% CI]	Recall (95% CI) [missed of 24]	Specificity (95% CI)	F1	NPV	Accuracy
Primary pipeline (pre-specified ensemble)	0.862 [0.765, 0.936]	1.00 (0.86–1.00) [0]	0.00 (0.00–0.03)	0.31	n/a	0.19
Member: engineered, pooled LR	0.832 [0.729, 0.916]	0.96 (0.79–1.00) [1]	0.18 (0.11–0.27)	0.35	0.95	0.33
Member: engineered, attention MIL	0.705 [0.578, 0.819]	1.00 (0.86–1.00) [0]	0.00 (0.00–0.03)	0.31	n/a	0.19
Member: Macenko + fine-tuned CNN	0.737 [0.629, 0.832]	1.00 (0.86–1.00) [0]	0.07 (0.03–0.13)	0.33	1.00	0.24
Secondary pipeline (Macenko + frozen CNN + PCA-64)	0.660 [0.535, 0.770]	0.75 (0.53–0.90) [6]	0.47 (0.37–0.57)	0.37	0.89	0.52
Hematology foundation model (added in this revision, selected on C-NMC development data)						
DinoBloom-S, pooled LR	0.840 [0.732, 0.933]	1.00 (0.86–1.00) [0]	0.04 (0.01–0.09)	0.32	1.00	0.22
DinoBloom-S + engineered, pooled LR	0.877 [0.773, 0.960]	1.00 (0.86–1.00) [0]	0.05 (0.02–0.11)	0.32	1.00	0.22

TABLE XV

CAN THE RANKING REDUCE THE REVIEW WORKLOAD? PATIENTS ARE REFERRED FOR URGENT REVIEW IN ORDER OF THEIR SCORE. RECALL: FRACTION OF APL PATIENTS AMONG THE REFERRED ONES CAUGHT WHEN THE TOP 10%, 20% OR 30% OF PATIENTS ARE REFERRED (PATIENT-BOOTSTRAP 95% CI); PPV AT 10% APL PREVALENCE FOR THE SAME OPERATING POINT; WORKLOAD NEEDED TO CATCH 95% AND ALL APL PATIENTS (BOOTSTRAP 95% CI FOR 95%); AP: AVERAGE PRECISION (BASE RATE: 0.19 IN THE INDEPENDENT COHORT, 0.50 IN THE LATER COHORT). NO THRESHOLD IS FITTED ON THESE COHORTS.

			Recall when referring the top			PPV at 10%	Workload to catch	
Cohort	Model	AP [95% CI]	10%	20%	30%	prevalence (top 20%)	95% of APL	all APL
Independent (MLL) (24/129)	Primary pipeline	0.58 [0.40, 0.79]	0.333 [0.23, 0.52]	0.625 [0.44, 0.80]	0.792 [0.64, 0.95]	0.40	60% [29%, 93%]	90%
	DinoBloom + engineered	0.74 [0.56, 0.90]	0.458 [0.32, 0.65]	0.708 [0.56, 0.88]	0.792 [0.67, 0.96]	0.48	75% [28%, 84%]	79%
	Engineered, pooled LR	0.53 [0.35, 0.74]	0.333 [0.17, 0.47]	0.583 [0.40, 0.78]	0.750 [0.60, 0.92]	0.36	61% [33%, 95%]	93%
	Fine-tuned CNN	0.36 [0.23, 0.56]	0.208 [0.07, 0.35]	0.375 [0.21, 0.55]	0.542 [0.35, 0.77]	0.20	78% [45%, 88%]	84%
APL-AML later cohort (12/24)	Primary pipeline	0.82 [0.61, 0.97]	0.250 [0.12, 0.38]	0.333 [0.20, 0.56]	0.500 [0.33, 0.75]	0.31	75% [54%, 92%]	75%
	DinoBloom + engineered	0.81 [0.59, 0.98]	0.167 [0.08, 0.38]	0.333 [0.18, 0.56]	0.583 [0.33, 0.78]	0.31	71% [50%, 88%]	71%
	DinoBloom, attention MIL	0.84 [0.60, 0.99]	0.167 [0.07, 0.38]	0.333 [0.18, 0.57]	0.583 [0.38, 0.86]	0.31	67% [46%, 83%]	67%

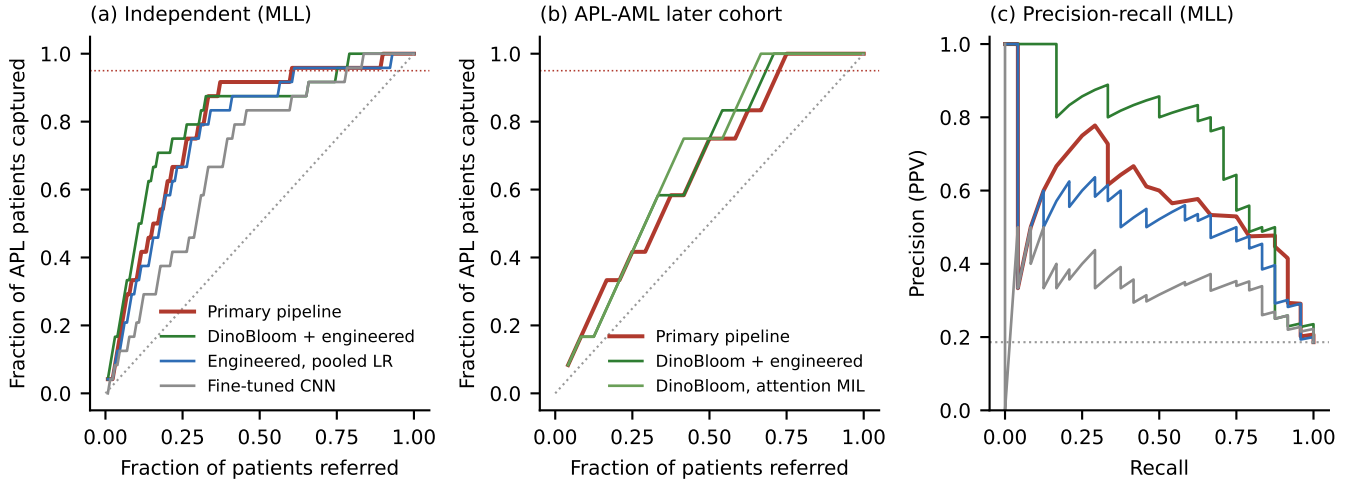


Fig. 7. Workload view of the ranking (no threshold fitted on the test cohorts). (a), (b) Fraction of APL patients caught versus the fraction of patients referred for urgent review, in score order; dotted diagonal: random referral; horizontal line: recall 0.95. (c) Precision–recall curves in the independent cohort; dotted: base rate 0.19.

in the later cohort [23] and compared their model with ten clinicians on the same 24 patients, finding it equivalent or better (reported only graphically); Manescu *et al.* reported 0.96 with random three-fold CV [4]. Neither reported results at a fixed recall. Re-run on the identical split with our operating-point rule, the promyelocyte proportion reached an AUROC of 0.611 and the full differential count 0.681; averaging single-cell predictions per patient [23] reached 0.681–0.799 and the CNN-feature-plus-SVM design of [16] 0.806. The primary pipeline reached 0.806 and missed no APL patient. With 24 external patients the confidence intervals of all learned models overlap; the clearest differences are over the manual biomarkers, in the number of missed patients, and between random CV (0.96) and temporal validation.

TABLE XVI

RECIPROCAL CROSS-LABORATORY TRANSFER AT PATIENT LEVEL. FORWARD: TRAINED ON APL-AML DISCOVERY (82 PATIENTS), TESTED ON THE INDEPENDENT MLL COHORT (24 APL, 105 AML). REVERSE: TRAINED ON THE MLL COHORT (129 PATIENTS; SETTINGS AND THRESHOLD FROM PATIENT-GROUPED CV WITHIN MLL), TESTED ON ALL 106 APL-AML PATIENTS (34 APL). DEV.: DEVELOPMENT CV AUROC. DISC./VAL.: REVERSE AUROC WITHIN THE EARLIER (82) AND LATER (24) APL-AML PATIENTS. RECALL AND SPECIFICITY AT THE SOURCE-COHORT THRESHOLD. THE APL-AML COHORT HAD BEEN EXAMINED EXTENSIVELY DURING DEVELOPMENT, SO THE REVERSE DIRECTION IS EXPLORATORY.

	APL-AML → MLL		MLL → APL-AML (exploratory)			
Model	Dev.	External [95% CI]	Dev.	All 106 [95% CI]	Disc./Val.	Recall / Spec.
Engineered, pooled LR	0.867	0.832 [0.729, 0.916]	0.909	0.750 [0.645, 0.840]	0.75/0.71	0.47 / 0.83
Engineered, attention MIL	0.827	0.705 [0.578, 0.819]	0.901	0.540 [0.486, 0.604]	0.48/0.64	0.00 / 1.00
DinoBloom-S, pooled LR	0.880	0.840 [0.732, 0.933]	0.958	0.764 [0.665, 0.853]	0.79/0.61	0.12 / 0.99
DinoBloom-S + engineered, pooled LR	0.869	0.877 [0.773, 0.960]	0.939	0.776 [0.679, 0.860]	0.79/0.69	0.06 / 1.00

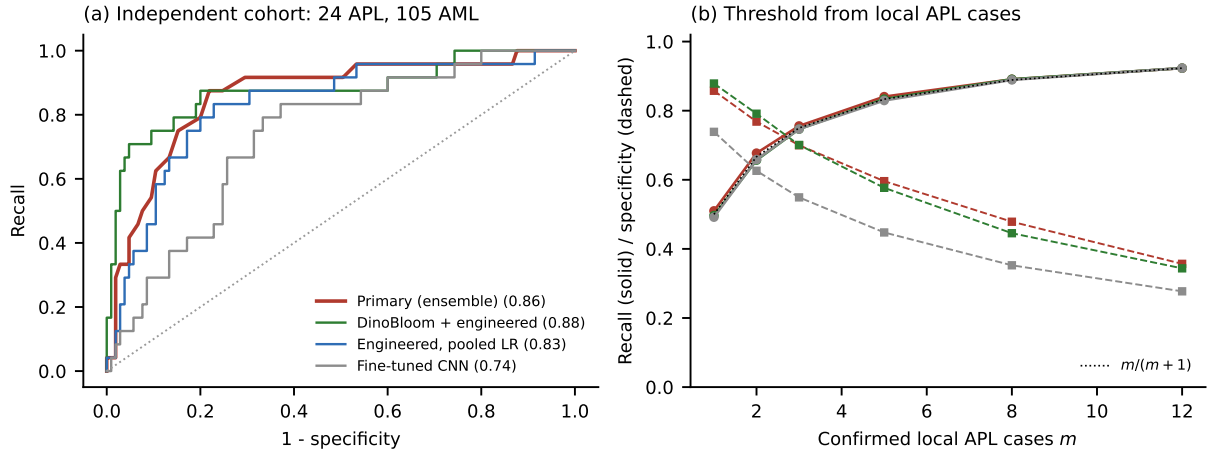


Fig. 8. Independent external test (24 APL, 105 other AML; models trained on APL-AML Discovery only). (a) ROC curves with AUROC in brackets. (b) Recall (solid) and specificity (dashed) when a new laboratory sets the threshold just below the lowest score of m confirmed local APL cases (2,000 random draws; dotted: $m/(m+1)$).

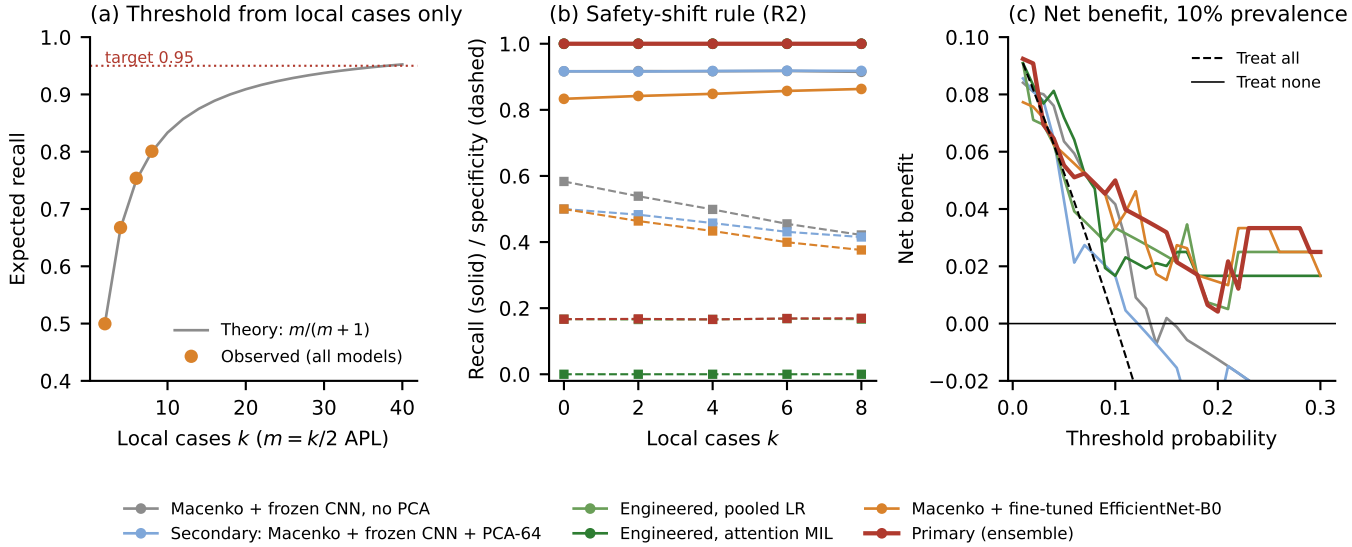


Fig. 9. Local threshold adaptation (APL-AML Validation cohort, exploratory). (a) Expected recall when the threshold is set just below the lowest score of $m = k/2$ local APL cases: observed values for $m \leq 4$ (the cohort's 12 APL patients allow no more) and the theoretical $m/(m+1)$, which beyond $m = 4$ is an extrapolation. (b) Recall (solid) and specificity (dashed) when the development threshold is lowered only as far as needed to catch the local APL cases. (c) Decision curves at 10% APL prevalence; dashed: test every patient; horizontal: test none.

V. DISCUSSION

Four findings stand out. First, how a model is evaluated changes the reported number more than how it is built, and

TABLE XVII
COMPARISON WITH EXISTING METHODS ON THE SAME APL-AML COHORT. UPPER BLOCK: RESULTS AS REPORTED BY THE ORIGINAL STUDIES, WHOSE PROTOCOLS DIFFER FROM OURS. LOWER BLOCKS: METHODS RE-RUN ON THE IDENTICAL SPLIT (TRAINED ON DISCOVERY, TESTED ONCE ON THE 24 VALIDATION PATIENTS) WITH THE SAME OPERATING-POINT RULE (SENSITIVITY ≥ 0.95 ON DISCOVERY). THE PRIMARY PIPELINE IS COMPARED WITH EACH OF ITS MEMBERS. RECALL AND SPECIFICITY WITH EXACT (CLOPPER–PEARSON) 95% CIS OVER 12 PATIENTS EACH; NPV N/A: NO PATIENT CALLED NEGATIVE.

Method	Evaluation	AUROC [95% CI]	Recall (95% CI) [missed]	Spec. (95% CI)	F1	NPV	Acc.
As reported in the literature							
Single-cell CNN, Sidhom *et al.* [23]	Temporal validation, single-cell classifier	0.743	–	–	–	–	–
Annotation-free MIL, Manescu *et al.* [4]	Random 3-fold CV, patient level	0.96 ± 0.02	–	–	–	–	–
Ten clinicians (reader study in [23])	Same 24 Validation patients, smear images	shown only graphically; the authors' MIL model was equivalent or better					
Re-run on the identical split (Validation cohort, patient level)							
Promyelocyte proportion (biomarker in [23])	Same protocol	0.611 [0.371, 0.837]	1.00 (0.74–1.00) [0]	0.00 (0.00–0.26)	0.67	n/a	0.50
Differential count + LR (comparator in [30])	Same protocol	0.681 [0.437, 0.886]	1.00 (0.74–1.00) [0]	0.08 (0.00–0.38)	0.69	1.00	0.54
CNN features + SVM (design of [16])	Same protocol	0.806 [0.615, 0.963]	0.50 (0.21–0.79) [6]	0.75 (0.43–0.95)	0.57	0.60	0.62
Single-cell classifier, engineered (design of [23])	Same protocol	0.681 [0.445, 0.908]	0.75 (0.43–0.95) [3]	0.50 (0.21–0.79)	0.67	0.67	0.62
Single-cell classifier, EfficientNet-B0 (design of [23])	Same protocol	0.799 [0.580, 0.972]	0.58 (0.28–0.85) [5]	0.92 (0.62–1.00)	0.70	0.69	0.75
Single-cell classifier, fusion (design of [23])	Same protocol	0.764 [0.536, 0.957]	0.75 (0.43–0.95) [3]	0.75 (0.43–0.95)	0.75	0.75	0.75
This work (pre-specified pipelines and their members)							
Primary pipeline (ensemble)	Same protocol	0.806 [0.615, 0.956]	1.00 (0.74–1.00) [0]	0.17 (0.02–0.48)	0.71	1.00	0.58
Member: engineered, pooled LR	Same protocol	0.757 [0.555, 0.938]	1.00 (0.74–1.00) [0]	0.17 (0.02–0.48)	0.71	1.00	0.58
Member: engineered, attention MIL	Same protocol	0.812 [0.614, 0.963]	1.00 (0.74–1.00) [0]	0.00 (0.00–0.26)	0.67	n/a	0.50
Member: Macenko + fine-tuned CNN	Same protocol	0.771 [0.563, 0.941]	0.83 (0.52–0.98) [2]	0.50 (0.21–0.79)	0.71	0.75	0.67
Secondary pipeline (Macenko + frozen CNN + PCA-64)	Same protocol	0.639 [0.400, 0.867]	0.92 (0.62–1.00) [1]	0.50 (0.21–0.79)	0.76	0.86	0.71

the overstatement has two separable sources. Patient leakage inflated AUROC by up to 0.11 when cells carried their patient's diagnosis, but much less when the label described the cell; the external generalization gap to a later cohort added up to 0.24. Leaky training did not make the models worse rankers, but it made their thresholds wrong. A study that reports image-level CV therefore overstates both the ranking and, more seriously, the recall that can be expected in practice. This held on two datasets and may explain why subject-disjoint challenge entries report F1 scores near 0.89 rather than near 1.0 [13], [14].

Second, the temporal APL-AML split carries a confounder that a patient-grouped CV within the development cohort cannot remove: APL patients were scanned earlier than most AML patients. Restricting the Discovery cohort to the period in which both diagnoses were scanned left the diagnostic performance of engineered morphology features unaffected—although these features carry the most scan-time information of all (0.794 among AML patients)—but cost the frozen CNN 0.10 AUROC, which suggests that generic deep features partly learn when a smear was scanned, and that part of the large external generalization gap on the temporal cohort may reflect the loss of this time-linked signal rather than a change in disease appearance; because the proxy is indirect, this interpretation is suggestive, not causal. Studies on this cohort, ours included, should report how their results depend on scan time.

Third, design choices followed pre-stated rules applied to the data rather than defaults. A label-free test for whether cells can be traced to their set found a stain shift in APL-AML and none in C-NMC, and this one observation explained several results: stain normalization helped modestly on APL-AML (the primary pipeline ranked 0.02 AUROC better with it, at the same operating point) and slightly hurt on C-NMC, development thresholds transferred to C-NMC but not to the later APL cohort, and PCA interacted with normalization. Fine-tuning and, more clearly, a hematology foundation model were the representation choices that helped on every test set; nested early stopping, attention pooling, cropping and gradient boosting did not improve on simpler settings, and PCA traded a little performance for a much smaller model. Across laboratories, interpretable engineered features transferred better than ImageNet networks, even after stain normalization, and combining them with DinoBloom gave the best ranking on the independent cohort.

Fourth, discrimination, threshold transport and clinical workload are separate questions, and only the first was answered favorably. Trained on 82 patients from one hospital, the pre-specified pipeline ranked APL in 129 patients from another laboratory with an AUROC of 0.86, higher than on the later cohort of its own hospital, and ranking also transferred in the reverse direction. But its development threshold flagged every one of the 105 AML patients in the independent cohort: 100% sensitivity at 0% specificity is not a calibration detail to be explained away, it is the central result. Used as a ranking, the model would still require review of 60% of patients to

capture 95% of APL patients. A threshold set from m local positives has an expected recall of only $m/(m+1)$, varies widely between laboratories, and says nothing about the false-positive burden; and validating a locked threshold requires separate positive cases—under binomial assumptions, all of 59 must be detected for a one-sided 95% lower bound of 0.95 on sensitivity— and 24 external patients are far below the sample sizes recommended for external validation [48], [49].

Relation to recent work. That patient-level splitting matters for C-NMC has been shown before [17]; our C-NMC analysis is a replication, and the contribution lies in the APL analyses: the separation of leakage from the external generalization gap on a temporal split, the scan-time confounding, the prespecified cross-laboratory evaluation, and the evaluation of thresholds and workload. cAItomorph [30] differs in three ways that make its higher AUROCs not comparable: it was developed on data collected from 6,610 individuals at the Munich Leukemia Laboratory, the institution of our independent cohort though from a different collection period, whereas our models never saw that laboratory; it predicts broad hematological disease classes and malignancy, not APL among other AML subtypes; and it reports discrimination and triage at its own operating points, not threshold transport from another site. Its external evaluations include the cohort of Hehr *et al.* that we used; we do not know whether patients overlap, but for that model the cohort shares the source institution, whereas for ours it does not. Recent multicenter evaluations of blood-cell classification [52] likewise show that robustness across hospitals has become the relevant benchmark for hematology AI; our results add that robust ranking and a transportable decision rule must be shown separately.

Limitations. C-NMC differs in disease (ALL vs. normal B cells), so it tests the evaluation principles rather than the APL model; the independent MLL cohort tests the APL model itself, but with a different negative population (genetically defined AML subtypes) and without the healthy donors. The external sets are small, and the APL-AML Validation cohort was evaluated repeatedly during development and is best regarded as a reused test set (Section III-F). The scan-time proxy is indirect. Cell labels in APL-AML come from an analyzer verified by an operator, and our cytoplasm features rely on proxies. Fine-tuning was partial, limited to 5,000 cells per fold and run with one seed on a CPU, and DinoBloom was used frozen and in its smallest version; stain augmentation [31], full fine-tuning and larger foundation models may perform better. No prospective or reader-in-the-loop evaluation was performed.

VI. Conclusion

Reported accuracy for leukemia detection from blood smears depends strongly on how the evaluation is set up: splitting cells instead of patients inflated performance and, above all, the expected recall, and a temporal external generalization gap and a scan-time confounder accounted for most of the remaining optimism on the APL cohort. Patient-grouped evaluation and external validation are necessary but not sufficient for clinical translation of blood-smear leukemia classifiers. In this study, models retained moderate-to-good patient-ranking performance across time and laboratories (AUROC 0.81–0.88 for APL), whereas thresholds selected during development failed to maintain an acceptable false-positive burden, and the ranking alone reduced the review workload only modestly at a high sensitivity target. The findings support these methods as candidates for sensitivity-oriented prioritization research, and show the need for independent local calibration on positive and negative cases, workload-aware evaluation and prospective validation before clinical deployment. We recommend that studies in this area split data by patient, separate leakage from the external generalization gap, test for acquisition shift and confounding, and report discrimination, fixed-threshold performance and workload separately.

Declaration of Generative AI Use

During this work the authors used an AI assistant (Claude, Anthropic) to write and run analysis code, to prepare figures and tables, and to draft and edit the manuscript text, under the authors' direction. The authors reviewed and verified the code, the results and the text, and take full responsibility for the content of this publication.

Data and Code Availability

The APL-AML cohort is available at https://github.com/sidhomj/DeepAPL; C-NMC 2019 [32] and AML-Cytomorphology_MLL_Helmholtz [20] from The Cancer Imaging Archive; DinoBloom weights from the authors' Zenodo release [25]. Code, configuration files, the commands that produced every result (`reproduce.sh`), locked patient-level split files, patient-level predictions of all reported pipelines on every external cohort, both pre-specifications and the analysis registry are available at https://github.com/Sherif-Hussein/leukemia-leakage-benchmark The code, configurations and results used for this paper are those of commit `739c33f8f083e6836a42e234395baf43630a6763` (branch `paper-v4`); `PROVENANCE.md` lists, with commit identifiers and times, when each analysis was added and what each test set had been exposed to, `verify_claims.py` recomputes the main external results from the released predictions, and a clean re-run of all analyses on the independent cohort and of the reverse transfer reproduced all 53 compared numbers exactly (`REPRODUCIBILITY.md`).

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